

GET 311

Engineering Mathematics III

Worked Solutions

Application questions for the eight engineering departments

3 Units · Compulsory · LH 45
128 questions · 8 topics · 8 departments

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How to Use These Solutions

Each solution sits under the question it answers, in the same order as the question set: eight topics, and within each topic the eight departments with two questions each.

Question. The question is repeated in a blue box like this one.

The working follows, step by step. All mathematics is set with Word's built-in equation editor, so any equation can be clicked and edited.

Answer. The key results are collected in a green box like this one at the end of each solution.

Numerical answers were checked with a computer algebra system (SymPy). Where a question as written contains an inconsistency, the solution says so and shows the corrected working. Try each question yourself before reading its solution.

Topic 1 Solutions: Linear Algebra

1. Agricultural Engineering

Question 1: Independence and span of the fertilizer vectors

Question. A fertilizer blend is made from three stock fertilizers with N-P-K compositions $\mathbf{a} = (20, 10, 10)$, $\mathbf{b} = (0, 20, 20)$ and $\mathbf{c} = (10, 0, 30)$ (percent by mass). Decide whether these three vectors are linearly independent. Can any target composition in $\mathbb{R}^3$ be reached by blending them? Explain using the idea of span.

Put $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ as the columns of a matrix and take its determinant:

$$\det \begin{bmatrix} 20 & 0 & 10 \\ 10 & 20 & 0 \\ 10 & 20 & 30 \end{bmatrix} = 20(600 - 0) - 0 + 10(200 - 200) = 12\,000$$

Since the determinant is not zero, the three vectors are **linearly independent**. Three independent vectors in $\mathbb{R}^3$ form a basis, so they **span** $\mathbb{R}^3$: any composition (N, P, K) can be written as $\alpha \mathbf{a} + \beta \mathbf{b} + \gamma \mathbf{c}$.

In practice a blend needs $\alpha, \beta, \gamma \geq 0$, since you cannot add a negative amount of fertilizer. The compositions you can actually make are only those with non-negative coefficients, which form a cone inside $\mathbb{R}^3$.

Answer. $\det = 12\,000 \neq 0$, so the vectors are independent and span $\mathbb{R}^3$. Only non-negative combinations are physically possible.

Question 2: Distances and angle between soil samples

Question. Soil samples from a farm are described by the vector (moisture %, organic matter %, pH). Given $\mathbf{s}_1 = (22, 3.1, 6.2)$, $\mathbf{s}_2 = (18, 2.4, 5.8)$ and a reference sample $\mathbf{r} = (20, 2.8, 6.0)$, find the Euclidean distance of each sample from the reference and the angle between $\mathbf{s}_1$ and $\mathbf{s}_2$. Comment on how alike the samples are.

$$\|\mathbf{s}_1 - \mathbf{r}\| = \|(2, 0.3, 0.2)\| = \sqrt{4 + 0.09 + 0.04} = 2.03$$

$$\|\mathbf{s}_2 - \mathbf{r}\| = \|(-2, -0.4, -0.2)\| = \sqrt{4 + 0.16 + 0.04} = 2.05$$

$$\cos \theta = \frac{\mathbf{s}_1 \cdot \mathbf{s}_2}{\|\mathbf{s}_1\| \|\mathbf{s}_2\|} = \frac{396 + 7.44 + 35.96}{(23.07)(19.07)} = \frac{439.4}{439.9} = 0.9993 \Rightarrow \theta \approx 2.2^\circ$$

Both samples are about the same distance from the reference, on opposite sides of it. The very small angle means they have almost the same *proportions* and differ mainly in overall size, which here is driven by moisture. The distance is dominated by moisture because it has the largest numbers, so in practice each variable should be scaled (normalised) before comparing.

Answer. Distances 2.03 and 2.05; angle $\theta \approx 2.2^\circ$. The samples are very alike.

2. Chemical Engineering

Question 1: Balancing $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

Question. The combustion of methane, $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$, can be balanced by writing the atom counts (C, H, O) of each species as vectors and finding a non-trivial vector in the null space of the resulting matrix. Set up the vector equation and find the smallest integer stoichiometric coefficients.

Let the coefficients be x_1, x_2, x_3, x_4 and write each species as a (C, H, O) vector:

$$x_1 \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} - x_3 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - x_4 \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = \mathbf{0}$$

This gives $x_1 - x_3 = 0$, $4x_1 - 2x_4 = 0$ and $2x_2 - 2x_3 - x_4 = 0$. The null space is spanned by $(1/2, 1, 1/2, 1)$. Multiplying by 2 gives the smallest integers $(1, 2, 1, 2)$.

Answer. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

Question 2: The stream compositions form a basis

Question. In a three-component mixture, the composition vectors of streams A, B and C are $\mathbf{x}_A = (0.5, 0.3, 0.2)$, $\mathbf{x}_B = (0.2, 0.5, 0.3)$ and $\mathbf{x}_C = (0.3, 0.2, 0.5)$. Show that the set $\{\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C\}$ is a basis of $\mathbb{R}^3$, and express a product stream $\mathbf{x}_P = (0.35, 0.35, 0.30)$ as a linear combination of them.

$$\det \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.3 & 0.5 \end{bmatrix} = 0.07 \neq 0$$

so the three vectors are independent and form a basis of $\mathbb{R}^3$. Solving $\alpha \mathbf{x}_A + \beta \mathbf{x}_B + \gamma \mathbf{x}_C = (0.35, 0.35, 0.30)$ by Gaussian elimination gives

$$\alpha = \frac{3}{7} \approx 0.429, \beta = \frac{5}{14} \approx 0.357, \gamma = \frac{3}{14} \approx 0.214$$

As a check, $\alpha + \beta + \gamma = 1$, as it must be when mixing streams whose fractions each add to 1.

Answer. $\mathbf{x}_P = 3/7 \mathbf{x}_A + 5/14 \mathbf{x}_B + 3/14 \mathbf{x}_C$, that is 42.9 % A, 35.7 % B and 21.4 % C.

3. Computer Engineering

Question 1: Rotation followed by scaling

Question. A 2-D graphics engine rotates every point by $\theta = 30^\circ$ and then scales it by 2 in the x -direction. Write each step as a linear transformation matrix, find the combined matrix, and apply it to the vertices of the unit square.

$$R(30^\circ) = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix}, S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

The rotation is applied first, so the combined matrix is

$$M = S R = \begin{bmatrix} \sqrt{3} & -1 \\ 0.5 & 0.866 \end{bmatrix} \approx \begin{bmatrix} 1.732 & -1 \\ 0.5 & 0.866 \end{bmatrix}$$

Applying M to the vertices of the unit square:

Vertex	Image
(0,0)	(0,0)
(1,0)	(1.732, 0.5)
(1,1)	(0.732, 1.366)
(0,1)	(-1, 0.866)

Since $\det M = 2 \times 1 = 2$, the image is a parallelogram of area 2.

Answer. $M = \begin{bmatrix} \sqrt{3} & -1 \\ 0.5 & 0.866 \end{bmatrix}$; the square becomes a parallelogram of area 2.

Question 2: Cosine similarity

Question. In a basic document-search system, each document is a term-frequency vector. With $\mathbf{d}_1 = (3, 0, 1, 2)$ and $\mathbf{d}_2 = (1, 1, 0, 2)$, compute the cosine similarity between them. Write a short algorithm (pseudocode) that returns the document most similar to a query vector $\mathbf{q}$.

$$\cos \theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|} = \frac{3+0+0+4}{\sqrt{14} \sqrt{6}} = \frac{7}{\sqrt{84}} = 0.764$$

```
function most_similar(q, docs):
    best, best_score = None, -1
    for each d in docs:
        score = dot(q, d) / (norm(q) * norm(d))
        if score > best_score: best, best_score = d, score
    return best
```

Answer. The cosine similarity is 0.764.

4. Civil Engineering

Question 1: Equilibrium at the truss joint

Question. Three forces act at a truss joint: $\mathbf{F}_1 = (10, 0)$ kN, $\mathbf{F}_2 = (-5, 8)$ kN and $\mathbf{F}_3 = (a, b)$ kN. Use vector addition to find $\mathbf{F}_3$ so the joint is in equilibrium. Are $\mathbf{F}_1$ and $\mathbf{F}_2$ linearly independent? What does that mean for the joint?

For equilibrium, $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$, so

$$\mathbf{F}_3 = -(10 - 5, 0 + 8) = (-5, -8) \text{ kN}, \|\mathbf{F}_3\| = 9.43 \text{ kN}$$

$\mathbf{F}_1$ and $\mathbf{F}_2$ are linearly independent because $\det \begin{bmatrix} 10 & -5 \\ 0 & 8 \end{bmatrix} = 80 \neq 0$. They span the plane, so any load at the joint can be resolved uniquely into components along the two members. This is what makes the joint statically determinate.

Answer. $\mathbf{F}_3 = (-5, -8)$ kN. $\mathbf{F}_1$ and $\mathbf{F}_2$ are independent, so the member forces are unique.

Question 2: Survey vectors

Question. A surveyor records displacement vectors between control points: $\overrightarrow{AB} = (120, 45, 2)$ m and $\overrightarrow{BC} = (-30, 80, -1)$ m. Find $\overrightarrow{AC}$, its length, and the unit vector along $\overrightarrow{AC}$. Find the projection of $\overrightarrow{BC}$ on $\overrightarrow{AB}$.

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = (90, 125, 1) \text{ m}, \|\overrightarrow{AC}\| = \sqrt{8100 + 15625 + 1} = 154.03 \text{ m}$$

The unit vector along $\overrightarrow{AC}$ is $\hat{\mathbf{u}} = (0.584, 0.812, 0.0065)$.

$$\overrightarrow{BC} \cdot \overrightarrow{AB} = -3600 + 3600 - 2 = -2, \text{comp}_{AB} \overrightarrow{BC} = \frac{-2}{128.18} = -0.0156 \text{ m}$$

The vector projection is $\frac{-2}{16429} \overrightarrow{AB} \approx (-0.015, -0.005, -0.0002)$ m, so BC is almost exactly perpendicular to AB.

Answer. $\overrightarrow{AC} = (90, 125, 1)$ m, length 154.03 m; projection of BC on AB is -0.0156 m.

5. Electrical and Electronics Engineering

Question 1: The three-phase phasors

Question. A three-phase balanced voltage set can be written as vectors in $\mathbb{R}^2$ (phasor components). Show that the phasors $V \angle 0^\circ$, $V \angle -120^\circ$ and $V \angle 120^\circ$ are linearly dependent, and that their sum is the zero vector.

In components, $V \angle 0^\circ = V(1, 0)$, $V \angle -120^\circ = V(-1/2, -\sqrt{3}/2)$ and $V \angle 120^\circ = V(-1/2, \sqrt{3}/2)$. Their sum is

$$V(1 - 1/2 - 1/2, 0 - \sqrt{3}/2 + \sqrt{3}/2) = (0, 0)$$

Since $1 \cdot \mathbf{v}_1 + 1 \cdot \mathbf{v}_2 + 1 \cdot \mathbf{v}_3 = \mathbf{0}$ with coefficients that are not all zero, the set is **linearly dependent**. (Any three vectors in $\mathbb{R}^2$ are dependent anyway.) This is why a balanced three-phase system has zero neutral current.

Answer. The phasors sum to $\mathbf{0}$, so they are linearly dependent.

Question 2: Haar-type expansion

Question. A discrete signal of 4 samples, $\mathbf{x} = (4, 2, 0, 2)$, is expanded in the orthogonal basis $\mathbf{h}_1 = (1, 1, 1, 1)$, $\mathbf{h}_2 = (1, 1, -1, -1)$, $\mathbf{h}_3 = (1, -1, 0, 0)$, $\mathbf{h}_4 = (0, 0, 1, -1)$ (a Haar-type basis). Check that the basis is orthogonal, then find the coefficients of $\mathbf{x}$ using inner products.

Every pair of basis vectors has zero dot product, for example $\mathbf{h}_1 \cdot \mathbf{h}_2 = 1 + 1 - 1 - 1 = 0$, so the basis is orthogonal. The coefficients are

$$c_i = \frac{\mathbf{x} \cdot \mathbf{h}_i}{\mathbf{h}_i \cdot \mathbf{h}_i}: c_1 = \frac{8}{4} = 2, c_2 = \frac{4}{4} = 1, c_3 = \frac{2}{2} = 1, c_4 = \frac{-2}{2} = -1$$

Check: $2(1, 1, 1, 1) + (1, 1, -1, -1) + (1, -1, 0, 0) - (0, 0, 1, -1) = (4, 2, 0, 2) \checkmark$

Answer. $\mathbf{x} = 2\mathbf{h}_1 + \mathbf{h}_2 + \mathbf{h}_3 - \mathbf{h}_4$

6. Food Engineering

Question 1: Ingredient masses

Question. A food product is formulated from three ingredients whose (protein, fat, carbohydrate) contents in g per 100 g are $(12, 2, 70)$, $(25, 15, 5)$ and $(1, 0, 95)$. Decide whether any nutritional profile can be made by mixing them (span). Find the mass of each ingredient needed to supply exactly $(10, 3, 75)$ g of protein, fat and carbohydrate.

The three composition vectors are independent ($\det \neq 0$), so together they span $\mathbb{R}^3$. Let x, y, z be the masses in units of 100 g:

$$\begin{bmatrix} 12 & 25 & 1 \\ 2 & 15 & 0 \\ 70 & 5 & 95 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 75 \end{bmatrix} \Rightarrow x=0.532, y=0.129, z=0.391$$

Answer. 53.2 g, 12.9 g and 39.1 g of the three ingredients (105.2 g in total).

Question 2: Colour difference

Question. The colour of a juice is measured in (L^*, a^*, b^*) colour space. The target is $(45, 20, 30)$ and a batch measures $(42, 24, 27)$. Compute the colour difference ΔE as a Euclidean norm, and the unit vector giving the direction of the colour error.

$$\mathbf{e} = \text{batch} - \text{target} = (-3, 4, -3), \Delta E = \sqrt{9+16+9} = \sqrt{34} \approx 5.83$$

The unit error direction is $\hat{\mathbf{e}} = (-0.514, 0.686, -0.514)$. The batch is darker (lower L^*), redder (higher a^*) and less yellow (lower b^*) than the target. A ΔE above about 2–3 is visible to the eye.

Answer. $\Delta E = 5.83$; direction $(-0.514, 0.686, -0.514)$.

7. Mechanical Engineering

Question 1: Moment of the force

Question. A force $\mathbf{F} = (3, -2, 5)$ N acts at the point $\mathbf{r} = (1, 2, 0)$ m. Compute the moment $\mathbf{M} = \mathbf{r} \times \mathbf{F}$ about the origin, and check that $\mathbf{M}$ is orthogonal to both $\mathbf{r}$ and $\mathbf{F}$ using the dot product.

$$\mathbf{M} = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 0 \\ 3 & -2 & 5 \end{vmatrix} = (10-0)\mathbf{i} - (5-0)\mathbf{j} + (-2-6)\mathbf{k} = (10, -5, -8) \text{ N}\cdot\text{m}$$

Checks: $\mathbf{M} \cdot \mathbf{r} = 10 - 10 + 0 = 0$ ✓ and $\mathbf{M} \cdot \mathbf{F} = 30 + 10 - 40 = 0$ ✓, so $\mathbf{M}$ is orthogonal to both $\mathbf{r}$ and $\mathbf{F}$.

Answer. $\mathbf{M} = (10, -5, -8)$ N·m

Question 2: The deformation map

Question. The displacement field of a deformed plate is modelled by a linear map $T(x, y) = (x + 0.02y, 0.01x + y)$. Write the matrix of T , find the image of the unit vectors, and decide whether T is invertible.

$$T = \begin{bmatrix} 1 & 0.02 \\ 0.01 & 1 \end{bmatrix}, T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.01 \end{bmatrix}, T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.02 \\ 1 \end{bmatrix}$$

$\det T = 1 - 0.0002 = 0.9998 \neq 0$, so T is **invertible**. Area changes by only -0.02% , so the deformation is nearly area-preserving.

Answer. $\det T = 0.9998$, so T is invertible.

8. Petroleum Engineering

Question 1: The blend

Question. The composition of a reservoir fluid is recorded as mole fractions of (C_1, C_2, C_3, C_4) . Samples from wells W_1 and W_2 are $(0.70, 0.12, 0.08, 0.10)$ and $(0.60, 0.15, 0.10, 0.15)$. Find the composition of a blend with 40 % of W_1 and 60 % of W_2 , and the Euclidean distance between the two well compositions.

$$\mathbf{x} = 0.4 \mathbf{w}_1 + 0.6 \mathbf{w}_2 = (0.640, 0.138, 0.092, 0.130)$$

which sums to 1 ✓. The distance between the wells is

$$\|\mathbf{w}_1 - \mathbf{w}_2\| = \sqrt{0.01 + 0.0009 + 0.0004 + 0.0025} = 0.117$$

Answer. Blend $(0.640, 0.138, 0.092, 0.130)$; distance 0.117.

Question 2: The well path

Question. A deviated well path is given by survey displacement vectors $\mathbf{d}_1 = (0, 0, -500)$ m, $\mathbf{d}_2 = (150, 100, -400)$ m and $\mathbf{d}_3 = (300, 200, -200)$ m. Find the total displacement vector to the target and the horizontal departure. Decide whether the three vectors are linearly independent.

$$\mathbf{d}_1 + \mathbf{d}_2 + \mathbf{d}_3 = (450, 300, -1100) \text{ m, horizontal departure} = \sqrt{450^2 + 300^2} = 540.8 \text{ m}$$

$\det[\mathbf{d}_1 \ \mathbf{d}_2 \ \mathbf{d}_3] = 0$, so the vectors are **linearly dependent**; in fact $\mathbf{d}_3 = 2\mathbf{d}_2 - 1.2\mathbf{d}_1$. The three survey legs lie in a single vertical plane, with azimuth $\tan^{-1}(450/300) \approx 56.3^\circ$ from north if y points north.

Answer. Total displacement $(450, 300, -1100)$ m; departure 540.8 m; the vectors are dependent.

Topic 2 Solutions: Matrices, Determinants and Inverses

1. Agricultural Engineering

Question 1: Revenue per hectare

Question. A farm makes three crops (maize, cassava, yam) over two seasons. The yield matrix Y (tonnes/ha) is 2×3 and the price vector $\mathbf{p}$ (₦/tonne) is 3×1 . With $Y = \begin{bmatrix} 2.5 & 12 & 8 \\ 3.0 & 10 & 9 \end{bmatrix}$ and $\mathbf{p} = (150\,000, 40\,000, 60\,000)^T$, compute the revenue per hectare for each season using matrix multiplication.

$$\mathbf{R} = Y \mathbf{p} = \begin{bmatrix} 2.5 & 12 & 8 \\ 3.0 & 10 & 9 \end{bmatrix} \begin{bmatrix} 150\,000 \\ 40\,000 \\ 60\,000 \end{bmatrix} = \begin{bmatrix} 375\,000 + 480\,000 + 480\,000 \\ 450\,000 + 400\,000 + 540\,000 \end{bmatrix} = \begin{bmatrix} 1\,335\,000 \\ 1\,390\,000 \end{bmatrix}$$

Answer. Season 1: ₦1 335 000 per ha. Season 2: ₦1 390 000 per ha.

Question 2: The hitch linkage

Question. A tractor's hitch linkage gives the transformation matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ between input and output displacements. Find $\det A$ and A^{-1} , and explain what $\det A$ says about how the linkage changes the area of the motion region.

$$\det A = 2 \cdot 3 - 1 \cdot 1 = 5, A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix}$$

The determinant is the area scale factor. Any region of input motion is mapped to an output region **5 times larger**. Because $\det A > 0$, orientation is kept.

Answer. $\det A = 5$; the linkage multiplies areas by 5.

2. Chemical Engineering

Question 1: Feed flow rates

Question. A mixer takes three feed streams, and the composition matrix C (rows = components, columns = streams) is $C = \begin{bmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.3 & 0.7 \end{bmatrix}$. Find C^{-1} and use it to get the feed flow rates that give component product flows $(50, 40, 60)$ kg/h.

$\det C = 0.14 \neq 0$, so C is invertible, and

$$C^{-1} = \frac{1}{14} \begin{bmatrix} 29 & -11 & -1 \\ -19 & 41 & -9 \\ 4 & -16 & 24 \end{bmatrix}, \mathbf{F} = C^{-1} \begin{bmatrix} 50 \\ 40 \\ 60 \end{bmatrix} = \begin{bmatrix} 67.86 \\ 10.71 \\ 71.43 \end{bmatrix} \text{ kg/h}$$

Check, component 1: $0.6(67.86) + 0.2(10.71) + 0.1(71.43) = 50.0 \checkmark$

Answer. $F_1 = 67.86 \text{ kg/h}$, $F_2 = 10.71 \text{ kg/h}$, $F_3 = 71.43 \text{ kg/h}$.

Question 2: Rows that sum to zero

Question. The rate constants of a reversible reaction network are placed in a 3×3 matrix K . Show that if the rows of K add to zero (mass conservation), then $\det K = 0$. Explain why this makes K singular in physical terms.

Let $\mathbf{1} = (1, 1, 1)^T$. If every row of K sums to zero, then

$$K \mathbf{1} = \mathbf{0}$$

so K has a non-zero vector in its null space, its columns are dependent, and $\det K = 0$.

Physically, the reactions only move mass between species. The total amount is conserved, so one combination of concentrations never changes and cannot be solved for independently. This also means there is always a non-trivial equilibrium (steady-state) composition.

Answer. $K \mathbf{1} = \mathbf{0}$ with $\mathbf{1} \neq \mathbf{0}$, so $\det K = 0$: conservation of mass removes one degree of freedom.

3. Computer Engineering

Question 1: Hill cipher key

Question. In a Hill cipher, a message is encrypted with the key matrix $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \pmod{26}$. Find $\det K \pmod{26}$, check that K is invertible mod 26, and find the decryption matrix $K^{-1} \pmod{26}$.

$\det K = 3 \cdot 5 - 3 \cdot 2 = 9$, and $\gcd(9, 26) = 1$, so K is invertible mod 26. The inverse of 9 mod 26 is 3, since $9 \cdot 3 = 27 \equiv 1$. Then

$$K^{-1} \equiv 3 \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} \equiv \begin{bmatrix} 15 & -9 \\ -6 & 9 \end{bmatrix} \equiv \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \pmod{26}$$

Check: $K K^{-1} = \begin{bmatrix} 105 & 78 \\ 130 & 79 \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \pmod{26} \checkmark$

Answer. $\det K \equiv 9 \pmod{26}$; $K^{-1} \equiv \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \pmod{26}$.

Question 2: Translate then rotate in homogeneous coordinates

Question. Homogeneous coordinates use 3×3 matrices for 2-D translation, rotation and scaling. Write the matrix that translates by $(4, -2)$ and then rotates by 90° . Find its determinant and inverse, and explain why the determinant is 1.

$$T = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}, R(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = RT = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 4 \\ 0 & 0 & 1 \end{bmatrix}, M^{-1} = \begin{bmatrix} 0 & 1 & -4 \\ -1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

M^{-1} rotates by -90° and then translates by $(-4, 2)$. The determinant is 1 because $\det R = 1$ (a rotation) and $\det T = 1$ (a translation); neither changes area, so their product doesn't either.

Answer. $\det M = 1$; M^{-1} is given above.

4. Civil Engineering

Question 1: Flexibility matrix and displacements

Question. A two-member spring model of a structure has stiffness matrix $K = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}$ with $k_1 = 200 \text{ kN/m}$ and $k_2 = 150 \text{ kN/m}$. Find K^{-1} (the flexibility matrix) and the displacements under loads $\mathbf{F} = (10, 5)^T \text{ kN}$.

$$K = \begin{bmatrix} 350 & -150 \\ -150 & 150 \end{bmatrix} \text{ kN/m}, \det K = 52500 - 22500 = 30000$$

$$K^{-1} = \frac{1}{30000} \begin{bmatrix} 150 & 150 \\ 150 & 350 \end{bmatrix} = \begin{bmatrix} 1/200 & 1/200 \\ 1/200 & 7/600 \end{bmatrix} \text{ m/kN}$$

$$\mathbf{u} = K^{-1} \mathbf{F} : u_1 = \frac{10+5}{200} = 0.075 \text{ m}, u_2 = \frac{10}{200} + \frac{35}{600} = 0.108 \text{ m}$$

Answer. $u_1 = 75 \text{ mm}$ and $u_2 = 108 \text{ mm}$.

Question 2: The unsupported beam element

Question. Show that the stiffness matrix of an unsupported beam element has determinant zero. Explain, using rank, why supports (boundary conditions) must be applied before the system can be solved.

The Euler-Bernoulli beam element stiffness matrix is

$$K = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

$R_1 + R_3 = \mathbf{0}$ and $R_2 + R_4 = L R_1$, where R_i is row i . So only two rows are independent: $\text{rank } K = 2 < 4$ and $\det K = 0$.

The two missing ranks are the two rigid-body motions: translating and rotating the whole beam produces no strain and no force. With K singular, $K \mathbf{u} = \mathbf{F}$ has no unique solution. Supports remove these rigid-body modes, which removes the matching rows and columns, and the reduced matrix is non-singular.

Answer. $\text{rank } K = 2$, so $\det K = 0$; supports must remove the two rigid-body modes.

5. Electrical and Electronics Engineering

Question 1: Admittance matrix and port currents

Question. A two-port network has the impedance matrix $Z = \begin{bmatrix} 10 & 4 \\ 4 & 6 \end{bmatrix} \Omega$. Find the admittance matrix $Y = Z^{-1}$ and the port currents when $\mathbf{V} = (20, 12)^T \text{ V}$.

$$\det Z = 60 - 16 = 44, Y = Z^{-1} = \frac{1}{44} \begin{bmatrix} 6 & -4 \\ -4 & 10 \end{bmatrix} = \begin{bmatrix} 0.136 & -0.091 \\ -0.091 & 0.227 \end{bmatrix} \text{ S}$$

$$I_1 = \frac{6(20) - 4(12)}{44} = \frac{72}{44} = 1.636 \text{ A}, I_2 = \frac{-4(20) + 10(12)}{44} = \frac{40}{44} = 0.909 \text{ A}$$

Answer. $I_1 = 1.636 \text{ A}$ and $I_2 = 0.909 \text{ A}$.

Question 2: Determinant of the conductance matrix

Question. Nodal analysis of a three-node resistive circuit gives the conductance matrix

$G = \begin{bmatrix} 0.5 & -0.2 & 0 \\ -0.2 & 0.7 & -0.3 \\ 0 & -0.3 & 0.4 \end{bmatrix} \text{ S}$. Compute $\det G$ by cofactor expansion and state whether a unique set of node voltages exists.

Expanding along the first row:

$$\det G = 0.5(0.7 \cdot 0.4 - 0.3 \cdot 0.3) + 0.2(-0.2 \cdot 0.4 - 0) + 0 = 0.095 - 0.016 = 0.079 \text{ S}^3$$

Answer. $\det G = 0.079 \neq 0$, so the node voltages are unique.

6. Food Engineering

Question 1: Batches that use all the stock

Question. A bakery makes bread, cake and biscuits. The ingredient-use matrix (kg of flour, sugar

and fat per batch) is $A = \begin{bmatrix} 5 & 3 & 2 \\ 0.5 & 2 & 1 \\ 0.3 & 1.5 & 1 \end{bmatrix}$. With daily stock $(80, 25, 18) \text{ kg}$, find A^{-1} and the

number of batches that uses all the stock exactly.

Rows are ingredients (flour, sugar, fat) and columns are products (bread, cake, biscuit). $\det A = 2.2$ and

$$A^{-1} = \begin{bmatrix} 5/22 & 0 & -5/11 \\ -1/11 & 2 & -20/11 \\ 3/44 & -3 & 85/22 \end{bmatrix}, \mathbf{x} = A^{-1} \begin{bmatrix} 80 \\ 25 \\ 18 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \\ 0 \end{bmatrix}$$

Check: flour $50 + 30 = 80 \checkmark$, sugar $5 + 20 = 25 \checkmark$, fat $3 + 15 = 18 \checkmark$

Answer. 10 batches of bread, 10 batches of cake and no biscuits.

Question 2: The dryer heat-transfer matrix

Question. The heat-transfer coefficients between three zones of a tunnel dryer form a symmetric matrix H . Given $H = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix}$, show that H is symmetric, compute $\det H$, and find H^{-1} .

$H^T = H$, since $h_{12} = h_{21} = -1$, $h_{23} = h_{32} = -1$ and $h_{13} = h_{31} = 0$, so H is symmetric.

$$\det H = 4(16 - 1) + 1(-4 - 0) + 0 = 56, H^{-1} = \frac{1}{56} \begin{bmatrix} 15 & 4 & 1 \\ 4 & 16 & 4 \\ 1 & 4 & 15 \end{bmatrix}$$

The inverse is symmetric too: the inverse of a symmetric matrix is always symmetric.

Answer. $\det H = 56$; H^{-1} is given above.

7. Mechanical Engineering

Question 1: Properties of the rotation matrix

Question. The rotation matrix about the z -axis is $R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Prove that R is orthogonal ($R^T R = I$), that $\det R = 1$, and that $R^{-1} = R(-\theta)$.

Let $c = \cos \theta$ and $s = \sin \theta$. Then

$$R^T R = \begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c^2 + s^2 & 0 & 0 \\ 0 & s^2 + c^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Expanding along the third row, $\det R = 1 \cdot (c^2 + s^2) = 1$. From $R^T R = I$ we get $R^{-1} = R^T$. Since $\cos(-\theta) = c$ and $\sin(-\theta) = -s$, the transpose is exactly $R(-\theta)$. ■

Answer. $R^T R = I$, $\det R = 1$ and $R^{-1} = R^T = R(-\theta)$.

Question 2: $M^{-1}K$

Question. A two-degree-of-freedom spring-mass system has mass matrix $M = \text{diag}(2, 1)$ kg and stiffness matrix $K = \begin{bmatrix} 300 & -100 \\ -100 & 100 \end{bmatrix}$ N/m. Compute $M^{-1}K$, which is needed for the vibration analysis in Topic 3.

$$M^{-1} = \text{diag}(1/2, 1), M^{-1}K = \begin{bmatrix} 150 & -50 \\ -100 & 100 \end{bmatrix} \text{s}^{-2}$$

Its eigenvalues are the squared natural frequencies ω^2 (see Topic 3).

Answer. $M^{-1}K = \begin{bmatrix} 150 & -50 \\ -100 & 100 \end{bmatrix}$

8. Petroleum Engineering

Question 1: Block pressures

Question. A finite-difference model of pressure in a 1-D reservoir with three blocks gives the

transmissibility matrix $T = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. Find T^{-1} and the block pressures when

$T \mathbf{p} = (1500, 0, 1200)^T$ (suitably scaled).

$$\det T = 4, T^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}, \mathbf{p} = T^{-1} \begin{bmatrix} 1500 \\ 0 \\ 1200 \end{bmatrix} = \begin{bmatrix} 1425 \\ 1350 \\ 1275 \end{bmatrix} \text{ psi}$$

The pressure falls steadily from the high-pressure boundary block towards the lower one, as expected for steady 1-D flow.

Answer. $\mathbf{p} = (1425, 1350, 1275)$ psi

Question 2: Production-rate matrix

Question. Three wells produce oil, gas and water. The production-rate matrix is

$P = \begin{bmatrix} 500 & 200 & 50 \\ 300 & 400 & 100 \\ 200 & 100 & 300 \end{bmatrix}$ (units/day). Find $\det P$ and explain what a zero determinant would mean

for back-calculating each well's contribution from total separator readings.

$$\det P = 500(120\,000 - 10\,000) - 200(90\,000 - 20\,000) + 50(30\,000 - 80\,000) = 3.85 \times 10^7$$

Since $\det P \neq 0$, each well's contribution can be recovered uniquely from the separator totals. If $\det P = 0$, at least one well's output would be a linear combination of the others. Infinitely many allocations would then fit the same totals, and the contributions could not be identified without extra measurements, such as individual well tests.

Answer. $\det P = 3.85 \times 10^7 \neq 0$, so the allocation is unique.

Topic 3 Solutions: Linear Equations, Eigenvalues and Eigenvectors

1. Agricultural Engineering

Question 1: Canal flows

Question. An irrigation network has three canals with flows q_1, q_2, q_3 (m^3/s) that satisfy $q_1 + q_2 + q_3 = 12$, $2q_1 - q_2 = 3$ and $q_2 - q_3 = 1$. Use Gaussian elimination to find the flows, and check the result with Cramer's rule.

From $2q_1 - q_2 = 3$ we get $q_1 = (3 + q_2)/2$, and from $q_2 - q_3 = 1$ we get $q_3 = q_2 - 1$. Substituting into the first equation:

$$\frac{3+q_2}{2} + q_2 + q_2 - 1 = 12 \Rightarrow 2.5q_2 = 11.5 \Rightarrow q_2 = 4.6$$

Then $q_1 = 3.8$ and $q_3 = 3.6$.

Check with Cramer's rule. With $D = \det \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} = 5$:

$$q_1 = \frac{D_1}{D} = \frac{19}{5} = 3.8, q_2 = \frac{D_2}{D} = \frac{23}{5} = 4.6, q_3 = \frac{D_3}{D} = \frac{18}{5} = 3.6 \checkmark$$

Answer. $q_1 = 3.8 \text{ m}^3/\text{s}$, $q_2 = 4.6 \text{ m}^3/\text{s}$, $q_3 = 3.6 \text{ m}^3/\text{s}$.

Question 2: Leslie matrix

Question. A livestock population has calves, yearlings and adults. It is modelled by the Leslie matrix $L = \begin{bmatrix} 0 & 0 & 0.8 \\ 0.6 & 0 & 0 \\ 0 & 0.9 & 0.9 \end{bmatrix}$. Find the dominant eigenvalue and its eigenvector. What do they say about long-term herd growth and the herd's age structure?

Expanding along the first row:

$$\det(L - \lambda I) = -\lambda(\lambda^2 - 0.9\lambda) + 0.8(0.54) = -\lambda^3 + 0.9\lambda^2 + 0.432$$

Setting this to zero gives $\lambda^3 - 0.9\lambda^2 - 0.432 = 0$. Its real root is $\lambda_1 = 1.2$, since $1.728 - 1.296 - 0.432 = 0$. The other two roots are complex, $-0.15 \pm 0.58i$, with modulus $0.6 < 1.2$.

The eigenvector for $\lambda = 1.2$, normalised to sum to 1, is $(1/3, 1/6, 1/2)$.

Answer. $\lambda_1 = 1.2$: the herd grows by 20 % per time step, settling to about 33 % calves, 17 % yearlings and 50 % adults.

2. Chemical Engineering

Question 1: CSTRs with recycle

Question. A system of three CSTRs in series with recycle gives the linear equations $5C_1 - C_3 = 20$, $-4C_1 + 5C_2 = 0$ and $-4C_2 + 5C_3 = 0$ (concentrations in mol/L). Decide whether the system is consistent, then solve it using LU decomposition.

$\det A = 5(25) - 0 + (-1)(16) = 109 \neq 0$, so the system is consistent with a unique solution. The LU factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -0.8 & 1 & 0 \\ 0 & -0.8 & 1 \end{bmatrix}, U = \begin{bmatrix} 5 & 0 & -1 \\ 0 & 5 & -0.8 \\ 0 & 0 & 4.36 \end{bmatrix}$$

Forward substitution in $L\mathbf{y} = \mathbf{b}$ gives $\mathbf{y} = (20, 16, 12.8)$. Back substitution in $U\mathbf{x} = \mathbf{y}$ then gives

$$C_3 = \frac{12.8}{4.36} = 2.936, C_2 = \frac{16 + 0.8(2.936)}{5} = 3.670, C_1 = \frac{20 + 2.936}{5} = 4.587$$

Answer. $C_1 = 4.587$, $C_2 = 3.670$, $C_3 = 2.936$ mol/L.

Question 2: $A \rightleftharpoons B$ kinetics

Question. Two first-order reactions $A \rightleftharpoons B$ are modelled by $\frac{d\mathbf{x}}{dt} = K\mathbf{x}$ with $K = \begin{bmatrix} -3 & 1 \\ 3 & -1 \end{bmatrix}$. Find the eigenvalues and eigenvectors of K , and use them to write the general solution. Interpret the zero eigenvalue as the equilibrium composition.

$$\det(K - \lambda I) = (-3 - \lambda)(-1 - \lambda) - 3 = \lambda^2 + 4\lambda = 0 \Rightarrow \lambda_1 = 0, \lambda_2 = -4$$

For $\lambda_1 = 0$: $3x_1 = x_2$, so $\mathbf{v}_1 = (1, 3)$. For $\lambda_2 = -4$: $x_1 + x_2 = 0$, so $\mathbf{v}_2 = (-1, 1)$. The general solution is

$$\mathbf{x}(t) = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + c_2 e^{-4t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

As $t \rightarrow \infty$, $\mathbf{x} \rightarrow c_1(1, 3)$: the equilibrium, with $[B]/[A] = 3 = k_1/k_{-1}$. The zero eigenvalue means the total $A + B$ is conserved. The eigenvalue -4 sets how fast equilibrium is approached, with time constant $1/4$.

Answer. $\lambda = 0, -4$; the equilibrium ratio is $B : A = 3 : 1$.

3. Computer Engineering

Question 1: Gauss-Seidel

Question. Write an algorithm (pseudocode or Python) for Gauss-Seidel iteration, and use it for 3 iterations on $10x - y = 9$, $-x + 10y - 2z = 7$, $-2y + 10z = 6$. State the convergence condition (diagonal dominance) and check it.

```
function gauss_seidel(A, b, x, iterations):
    n = length(b)
    repeat iterations times:
        for i = 1..n:
```

```

s = b[i] - sum(A[i][j]*x[j] for j != i)
x[i] = s / A[i][i]          # uses the newest values at once
return x

```

The equations rearrange to $x = \frac{9+y}{10}$, $y = \frac{7+x+2z}{10}$ and $z = \frac{6+2y}{10}$. Starting from $(0, 0, 0)$:

Iteration	x	y	z
1	0.9000	0.7900	0.7580
2	0.9790	0.9495	0.7899
3	0.9950	0.9575	0.7915

The exact solution is $(0.9958, 0.9579, 0.7916)$. The matrix is **strictly diagonally dominant**, because $10 > 1$, $10 > 1+2$ and $10 > 2$ (in each row, the diagonal entry is larger than the sum of the absolute values of the other entries), so Gauss-Seidel is guaranteed to converge.

Answer. After 3 iterations, $(x, y, z) = (0.9950, 0.9575, 0.7915)$.

Question 2: PageRank

Question. A simplified PageRank for 3 web pages uses the column-stochastic link matrix

$M = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{bmatrix}$. Show that $\lambda = 1$ is an eigenvalue, and find the normalised eigenvector that gives the page ranks.

Every column of M sums to 1, so $M^T \mathbf{1} = \mathbf{1}$ and $\lambda = 1$ is an eigenvalue of M^T . M and M^T have the same eigenvalues, so $\lambda = 1$ is an eigenvalue of M . (The eigenvalues are $1, -1/2, -1/2$.) Solving $(M - I)\mathbf{r} = \mathbf{0}$ gives $\mathbf{r} \propto (1, 1, 1)$.

Answer. $\mathbf{r} = (1/3, 1/3, 1/3)$: all three pages have equal rank.

4. Civil Engineering

Question 1: Principal stresses

Question. The plane stress at a point is given by the tensor $\sigma = \begin{bmatrix} 80 & 30 \\ 30 & 20 \end{bmatrix}$ MPa. Find the principal stresses (eigenvalues) and principal directions (eigenvectors), and give the angle of the principal plane.

$$\det(\sigma - \lambda I) = (80 - \lambda)(20 - \lambda) - 900 = \lambda^2 - 100\lambda + 700 = 0 \Rightarrow \lambda = 50 \pm 30\sqrt{2}$$

$$\sigma_1 = 92.43 \text{ MPa}, \sigma_2 = 7.57 \text{ MPa}$$

The principal directions are $\mathbf{v}_1 = (1 + \sqrt{2}, 1)$, at 22.5° to the x-axis, and $\mathbf{v}_2 = (1 - \sqrt{2}, 1)$, at 112.5° . As a check,

$$\tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{60}{60} = 1 \Rightarrow \theta = 22.5^\circ \checkmark$$

Answer. $\sigma_1 = 92.43 \text{ MPa}$ at 22.5° ; $\sigma_2 = 7.57 \text{ MPa}$ at 112.5° .

Question 2: Three-storey shear building

Question. A three-storey shear building has mass matrix $M = mI$ and stiffness matrix

$K = k \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$. Set up the eigenvalue problem $(K - \omega^2 M)\phi = \mathbf{0}$, and find the natural frequencies in terms of k/m .

With $M = mI$, the problem becomes $\frac{K}{k}\phi = \frac{\omega^2 m}{k}\phi$. The eigenvalues of $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ are 0.198, 1.555 and 3.247, so

$$\omega_1 = 0.445\sqrt{k/m}, \omega_2 = 1.247\sqrt{k/m}, \omega_3 = 1.802\sqrt{k/m}$$

In the first mode all floors move the same way, with the largest movement at the top.

Answer. $\omega = 0.445, 1.247, 1.802$ times $\sqrt{k/m}$.

5. Electrical and Electronics Engineering

Question 1: Mesh currents by Cramer's rule

Question. Mesh analysis of a circuit gives $12I_1 - 4I_2 = 24$, $-4I_1 + 10I_2 - 2I_3 = 0$ and $-2I_2 + 8I_3 = -6$. Solve for the mesh currents with Cramer's rule.

$$D = \det \begin{bmatrix} 12 & -4 & 0 \\ -4 & 10 & -2 \\ 0 & -2 & 8 \end{bmatrix} = 12(76) + 4(-32) = 784$$

$$D_1 = \det \begin{bmatrix} 24 & -4 & 0 \\ 0 & 10 & -2 \\ -6 & -2 & 8 \end{bmatrix} = 1776, D_2 = \det \begin{bmatrix} 12 & 24 & 0 \\ -4 & 0 & -2 \\ 0 & -6 & 8 \end{bmatrix} = 624, D_3 = \det \begin{bmatrix} 12 & -4 & 24 \\ -4 & 10 & 0 \\ 0 & -2 & -6 \end{bmatrix} = -432$$

$$I_1 = \frac{1776}{784} = 2.265 \text{ A}, I_2 = \frac{624}{784} = 0.796 \text{ A}, I_3 = \frac{-432}{784} = -0.551 \text{ A}$$

The negative I_3 flows opposite to the assumed direction.

Answer. $I_1 = 2.265 \text{ A}, I_2 = 0.796 \text{ A}, I_3 = -0.551 \text{ A}$.

Question 2: RLC circuit

Question. The state-space model of an RLC circuit is $\dot{\mathbf{x}} = A\mathbf{x}$ with $A = \begin{bmatrix} 0 & 1 \\ -1/(LC) & -R/L \end{bmatrix}$. For $R = 4 \Omega$, $L = 1 \text{ H}$ and $C = 0.25 \text{ F}$, find the eigenvalues of A . Classify the circuit as over-, critically or under-damped, and say whether it is stable.

$$A = \begin{bmatrix} 0 & 1 \\ -4 & -4 \end{bmatrix}, \det(A - \lambda I) = \lambda^2 + 4\lambda + 4 = (\lambda + 2)^2 = 0 \Rightarrow \lambda = -2$$

Repeated real eigenvalues mean the circuit is **critically damped**; as a check, $R = 4$ equals $2\sqrt{L/C} = 2\sqrt{4} = 4$. Both eigenvalues have negative real part, so the circuit is **stable**.

Answer. $\lambda = -2$ (double): critically damped and stable.

6. Food Engineering

Question 1: Juice blend

Question. A juice blender mixes three concentrates to meet targets for sugar, acidity and vitamin C. This gives $0.1x + 0.2y + 0.15z = 15$, $0.02x + 0.01y + 0.03z = 2$ and $0.5x + 0.2y + 0.4z = 40$. Decide using ranks whether the targets can be met exactly, and solve the system.

$\det A = 0.00105 \neq 0$, so $\text{rank } A = \text{rank}[A | \mathbf{b}] = 3$, the number of unknowns. The targets can be met exactly, with a unique blend:

$$x = 47.62, y = 33.33, z = 23.81$$

All three are positive, so the blend is physically possible.

Answer. Unique blend: $x = 47.62, y = 33.33, z = 23.81$ units.

Question 2: Drying model

Question. The moisture transfer between the surface, middle and core layers of a drying grain

kernel is modelled by $\frac{d\mathbf{M}}{dt} = A\mathbf{M}$ with $A = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$. Find the eigenvalues of A and

explain how the smallest eigenvalue in magnitude controls the drying time.

The eigenvalues of A are $-0.198, -1.555$ and -3.247 , and the solution is a sum of terms $c_i \mathbf{v}_i e^{\lambda_i t}$.

The fast modes die out quickly. The mode with the smallest $|\lambda|$, 0.198 , decays most slowly and controls the drying time:

$$\tau = \frac{1}{0.198} \approx 5.05, t_{1\%} \approx 4.6\tau \approx 23 \text{ time units}$$

Answer. $\lambda = -0.198, -1.555, -3.247$; the slowest mode ($\tau \approx 5$) sets the drying time.

7. Mechanical Engineering

Question 1: Natural frequencies and mode shapes

Question. For the spring-mass system of Topic 2 ($M = \text{diag}(2, 1)$, $K = \begin{bmatrix} 300 & -100 \\ -100 & 100 \end{bmatrix}$), solve

$\det(K - \omega^2 M) = 0$ to get the two natural frequencies, and find and sketch the corresponding mode shapes.

$$\det(K - \omega^2 M) = (300 - 2\omega^2)(100 - \omega^2) - 10000 = 0 \Rightarrow \omega^4 - 250\omega^2 + 10000 = 0$$

This gives $\omega^2 = 50$ or 200 , so $\omega_1 = 7.07$ rad/s and $\omega_2 = 14.14$ rad/s.

- Mode 1: $(300 - 100)x_1 = 100x_2$, so $x_2 = 2x_1$ and $\boldsymbol{\phi}_1 = (1, 2)$. Both masses move in the same direction.

- Mode 2: $(300 - 400)x_1 = 100x_2$, so $x_2 = -x_1$ and $\phi_2 = (1, -1)$. The masses move in opposite directions.

Answer. $\omega_1 = 7.07$ rad/s with mode $(1, 2)$; $\omega_2 = 14.14$ rad/s with mode $(1, -1)$.

Question 2: Cayley-Hamilton

Question. Use the Cayley-Hamilton theorem to find A^{-1} and A^3 for the inertia-like matrix

$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. Check A^{-1} by direct computation.

The characteristic equation is $\lambda^2 - 7\lambda + 10 = 0$, so $A^2 - 7A + 10I = 0$. Multiplying by A^{-1} :

$$A^{-1} = \frac{7I - A}{10} = \begin{bmatrix} 0.3 & -0.1 \\ -0.2 & 0.4 \end{bmatrix}$$

Using $A^2 = 7A - 10I$:

$$A^3 = 7A^2 - 10A = 7(7A - 10I) - 10A = 39A - 70I = \begin{bmatrix} 86 & 39 \\ 78 & 47 \end{bmatrix}$$

Direct check: $\det A = 10$, so $A^{-1} = \frac{1}{10} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix} \checkmark$

Answer. $A^{-1} = \begin{bmatrix} 0.3 & -0.1 \\ -0.2 & 0.4 \end{bmatrix}$ and $A^3 = \begin{bmatrix} 86 & 39 \\ 78 & 47 \end{bmatrix}$.

8. Petroleum Engineering

Question 1: Compartment pressures

Question. The material balance for three reservoir compartments linked by transmissibility gives $3p_1 - p_2 = 4000$, $-p_1 + 3p_2 - p_3 = 1000$ and $-p_2 + 2p_3 = 2500$ (psi). Solve for the compartment pressures with Gauss-Jordan elimination.

$$\begin{bmatrix} 3 & -1 & 0 & 4000 \\ -1 & 3 & -1 & 1000 \\ 0 & -1 & 2 & 2500 \end{bmatrix} \xrightarrow{R_2 \rightarrow 3R_2 + R_1} \begin{bmatrix} 3 & -1 & 0 & 4000 \\ 0 & 8 & -3 & 7000 \\ 0 & -1 & 2 & 2500 \end{bmatrix} \xrightarrow{R_3 \rightarrow 8R_3 + R_2} \begin{bmatrix} 3 & -1 & 0 & 4000 \\ 0 & 8 & -3 & 7000 \\ 0 & 0 & 13 & 27000 \end{bmatrix}$$

The last column is the right-hand side. Back substitution gives $p_3 = 27000/13 = 2076.9$, then $p_2 = 1653.8$ and $p_1 = 1884.6$.

Answer. $p_1 = 1884.6$ psi, $p_2 = 1653.8$ psi, $p_3 = 2076.9$ psi.

Question 2: Principal permeabilities

Question. The permeability tensor of an anisotropic formation is $\mathbf{K} = \begin{bmatrix} 200 & 50 \\ 50 & 100 \end{bmatrix}$ mD. Find the principal permeabilities and the directions of maximum and minimum permeability, and explain how these directions affect horizontal well placement.

$$\det(\mathbf{K} - \lambda \mathbf{I}) = \lambda^2 - 300\lambda + 17500 = 0 \Rightarrow \lambda = 150 \pm 50\sqrt{2}$$

So $k_{max} = 220.7$ mD along $(1 + \sqrt{2}, 1)$, at 22.5° from the x -axis, and $k_{min} = 79.3$ mD at 112.5° .

Fluid moves most easily along k_{max} . To get the most flow into the wellbore, a horizontal well should be drilled **perpendicular to k_{max}** , that is along 112.5° , so it cuts across the high-permeability direction. Hydraulic fractures, by contrast, tend to follow the stress field, not permeability.

Answer. $k_{max} = 220.7$ mD at 22.5° , $k_{min} = 79.3$ mD at 112.5° ; drill the horizontal well along 112.5° .

Topic 4 Solutions: Analytical Geometry and Coordinate Systems

1. Agricultural Engineering

Question 1: Centre-pivot boundary

Question. A centre-pivot irrigation system covers a circle of radius 400 m centred at (500, 300) m in the farm's coordinate grid. Write the equation of the irrigated boundary, convert it to polar coordinates centred at the pivot, and find the irrigated area.

In Cartesian coordinates the boundary is

$$(x - 500)^2 + (y - 300)^2 = 400^2$$

Translate the origin to the pivot with $X = x - 500$, $Y = y - 300$, so $X^2 + Y^2 = 160\,000$. In polar coordinates about the pivot ($X = r \cos \theta$, $Y = r \sin \theta$) this becomes $r = 400$. Then

$$A = \int_0^{2\pi} \int_0^{400} r \, dr \, d\theta = \pi (400)^2 = 502\,655 \text{ m}^2$$

Answer. $r = 400$ about the pivot; area $502\,655 \text{ m}^2 \approx 50.3 \text{ ha}$.

Question 2: The silo

Question. A grain silo is a cylinder of radius 3 m and height 10 m with a conical roof of height 2 m. Write the equations of both surfaces in cylindrical coordinates, and find the total storage volume.

In cylindrical coordinates (r, θ, z):

- Cylinder wall: $r = 3$, $0 \leq z \leq 10$.
- Conical roof: it rises from $z = 10$ at $r = 3$ to $z = 12$ at $r = 0$, so $z = 12 - 2/3 r$, $0 \leq r \leq 3$.

$$V = \pi r^2 h + 1/3 \pi r^2 h_c = \pi(9)(10) + 1/3 \pi(9)(2) = 90\pi + 6\pi = 96\pi \approx 301.6 \text{ m}^3$$

Answer. Wall $r = 3$; roof $z = 12 - 2/3 r$; volume $96\pi \approx 301.6 \text{ m}^3$.

2. Chemical Engineering

Question 1: The spherical catalyst pellet

Question. A spherical catalyst pellet of radius $R = 5 \text{ mm}$ has a reactant concentration $C(\rho)$ that depends only on the radial distance. Write the position of a point on the pellet surface in spherical coordinates, and explain why spherical coordinates reduce the 3-D diffusion problem to one variable.

A point on the surface is $(\rho, \theta, \phi) = (5 \text{ mm}, \theta, \phi)$ for any $0 \leq \theta < 2\pi$ and $0 \leq \phi \leq \pi$, that is $x = 5 \sin \phi \cos \theta$, $y = 5 \sin \phi \sin \theta$, $z = 5 \cos \phi$.

If C depends only on ρ , then $\partial C / \partial \theta = \partial C / \partial \phi = 0$ and the Laplacian reduces to

$$\nabla^2 C = \frac{1}{\rho^2} \frac{d}{d\rho} \left(\rho^2 \frac{dC}{d\rho} \right)$$

So the 3-D diffusion-reaction PDE becomes **one ordinary differential equation in ρ** , where Cartesian coordinates would need all three variables together.

Answer. Surface: $\rho = 5$ mm. Spherical symmetry removes θ and ϕ , leaving an ODE in ρ .

Question 2: The elliptical tank

Question. The cross-section of an elliptical storage tank is $9x^2 + 4y^2 - 36x + 24y + 36 = 0$. Use a translation of axes to reduce it to standard form, and find the centre, semi-axes and cross-sectional area.

Complete the squares:

$$9(x^2 - 4x) + 4(y^2 + 6y) + 36 = 0 \Rightarrow 9(x - 2)^2 + 4(y + 3)^2 = 36$$

Translating with $X = x - 2$ and $Y = y + 3$ gives the standard form

$$\frac{X^2}{4} + \frac{Y^2}{9} = 1$$

Answer. Centre $(2, -3)$; semi-axes $a = 2$ (along x) and $b = 3$ (along y); area $\pi ab = 6\pi \approx 18.85$.

3. Computer Engineering

Question 1: Robot arm position

Question. A robot arm's end-effector is located by $(r, \theta, z) = (0.8 \text{ m}, 60^\circ, 0.5 \text{ m})$ in cylindrical coordinates. Convert this to Cartesian coordinates, and write a function (pseudocode) that converts between Cartesian, cylindrical and spherical coordinates in both directions.

$$x = 0.8 \cos 60^\circ = 0.400 \text{ m}, y = 0.8 \sin 60^\circ = 0.693 \text{ m}, z = 0.5 \text{ m}$$

```

cart_to_cyl(x,y,z):  return (sqrt(x^2+y^2), atan2(y,x), z)
cyl_to_cart(r,θ,z):  return (r cos θ, r sin θ, z)
cart_to_sph(x,y,z):  ρ = sqrt(x^2+y^2+z^2)
                    return (ρ, atan2(y,x), acos(z/ρ))
sph_to_cart(ρ,θ,φ):  return (ρ sin φ cos θ, ρ sin φ sin θ, ρ cos φ)
cyl_to_sph(r,θ,z):   return (sqrt(r^2+z^2), θ, atan2(r, z))
sph_to_cyl(ρ,θ,φ):   return (ρ sin φ, θ, ρ cos φ)

```

Use `atan2`, not `atan`, so that the angle ends up in the correct quadrant.

Answer. $(x, y, z) = (0.400, 0.693, 0.5) \text{ m}$.

Question 2: The lidar point

Question. A lidar sensor reports a point at range $\rho = 20$ m, azimuth $\theta = 45^\circ$ and elevation angle 30° (so the polar angle is $\phi = 60^\circ$). Convert it to Cartesian coordinates, then find the distance from the point to the ground plane $z = 0$ and to the plane $x + y + z = 10$.

With $\rho = 20$, $\theta = 45^\circ$ and $\phi = 60^\circ$:

$$x = 20 \sin 60^\circ \cos 45^\circ = 12.25, y = 12.25, z = 20 \cos 60^\circ = 10$$

(all in metres).

The distance to $z=0$ is 10 m. The distance to the plane $x+y+z=10$ is

$$d = \frac{|12.25+12.25+10-10|}{\sqrt{3}} = \frac{24.49}{\sqrt{3}} = 14.14 \text{ m}$$

Answer. Point $(12.25, 12.25, 10)$ m; 10 m above the ground; 14.14 m from the plane.

4. Civil Engineering

Question 1: The embankment plane

Question. A road designer needs the equation of the plane that contains the points $A(0, 0, 100)$, $B(200, 0, 104)$ and $C(0, 150, 97)$ (metres) on a graded embankment. Find the plane equation, the direction cosines of its normal, and the angle it makes with the horizontal.

$\vec{AB} = (200, 0, 4)$ and $\vec{AC} = (0, 150, -3)$, so the normal is

$$\mathbf{n} = \vec{AB} \times \vec{AC} = (-600, 600, 30000) \parallel (-1, 1, 50)$$

The plane through A is $-x + y + 50z = 5000$, or

$$z = 100 + \frac{x - y}{50}$$

The direction cosines of the normal are $\frac{(-1, 1, 50)}{\sqrt{2502}} = (-0.0200, 0.0200, 0.9996)$. The angle with the horizontal equals the angle between the normal and the z -axis: $\cos^{-1}(0.9996) = 1.62^\circ$.

Answer. $z = 100 + (x - y)/50$; direction cosines $(-0.020, 0.020, 0.9996)$; slope angle 1.62° .

Question 2: The parabolic arch

Question. The profile of a parabolic arch bridge passes through $(-30, 0)$, $(30, 0)$ and $(0, 15)$ m. Find its equation. Rotate the axes by 10° to model an inclined abutment, and write the transformed equation.

The arch is symmetric with vertex $(0, 15)$, so $y = 15 - kx^2$. At $(30, 0)$, $0 = 15 - 900k$, so $k = 1/60$:

$$y = 15 - \frac{x^2}{60}$$

Rotate the axes by $\alpha = 10^\circ$ using $x = X \cos \alpha - Y \sin \alpha$ and $y = X \sin \alpha + Y \cos \alpha$, with $\cos 10^\circ = 0.9848$ and $\sin 10^\circ = 0.1736$:

$$0.1736 X + 0.9848 Y = 15 - \frac{(0.9848 X - 0.1736 Y)^2}{60}$$

Expanded, this contains an XY term. That is expected: in the rotated frame the parabola's axis is no longer parallel to a coordinate axis.

Answer. $y = 15 - x^2/60$; the rotated form is given above.

5. Electrical and Electronics Engineering

Question 1: The dipole pattern surface

Question. The radiation pattern of a dipole antenna is proportional to $\sin \phi$ in spherical coordinates. Convert the pattern surface $\rho = \sin \phi$ to Cartesian form, identify it, and sketch it.

Multiply $\rho = \sin \phi$ by ρ to get $\rho^2 = \rho \sin \phi$. Since $\rho \sin \phi = \sqrt{x^2 + y^2}$:

$$x^2 + y^2 + z^2 = \sqrt{x^2 + y^2}$$

In any plane through the z -axis, with $r = \sqrt{x^2 + y^2}$, this is $(r - 1/2)^2 + z^2 = 1/4$: a circle of radius $1/2$ centred at $r = 1/2$ that touches the axis. Revolving it about the z -axis gives a **horn torus**, a doughnut with no hole. The dipole radiates most strongly broadside ($\phi = 90^\circ$) and not at all along its own axis.

Answer. $x^2 + y^2 + z^2 = \sqrt{x^2 + y^2}$, a horn torus.

Question 2: The coaxial dielectric

Question. A coaxial cable has an inner conductor of radius $a = 1$ mm and an outer conductor of radius $b = 4$ mm. Describe the dielectric region with inequalities in cylindrical coordinates, and find the cross-sectional area of the dielectric using polar coordinates.

The dielectric region is $1 \text{ mm} \leq r \leq 4 \text{ mm}$, $0 \leq \theta < 2\pi$, with z unrestricted along the cable.

$$A = \int_0^{2\pi} \int_1^4 r \, dr \, d\theta = 2\pi \cdot \frac{16 - 1}{2} = 15\pi \approx 47.1 \text{ mm}^2$$

Answer. $1 \leq r \leq 4$ mm; area $15\pi \approx 47.1 \text{ mm}^2$.

6. Food Engineering

Question 1: Slowest-heating point in the can

Question. A cylindrical can with radius 4 cm and height 11 cm is heated during sterilisation. The slowest-heating point is on the axis at mid-height. Give its cylindrical coordinates relative to the centre of the base, and find its distance from the rim edge.

On the axis at mid-height means $(r, \theta, z) = (0, \text{any } \theta, 5.5 \text{ cm})$; the angle is undefined on the axis. The rim edges are at $r = 4$ and $z = 0$ or 11 , so the distance is

$$d = \sqrt{4^2 + 5.5^2} = \sqrt{46.25} = 6.80 \text{ cm}$$

Answer. $(0, \theta, 5.5)$ cm; 6.80 cm from the rim.

Question 2: The egg

Question. An egg is approximated by the ellipsoid $\frac{x^2}{4} + \frac{y^2}{4} + \frac{z^2}{9} = 1$ (cm). Find its volume, and use a coordinate transformation (stretching) to map it onto a unit sphere. Explain why this simplifies heat-transfer analysis.

With semi-axes $a = b = 2$ cm and $c = 3$ cm,

$$V = \frac{4}{3} \pi a b c = \frac{4}{3} \pi (2)(2)(3) = 16 \pi \approx 50.3 \text{ cm}^3$$

The stretching $X = x/2$, $Y = y/2$, $Z = z/3$ maps the ellipsoid onto the unit sphere $X^2 + Y^2 + Z^2 = 1$. The Jacobian is $abc = 12$, so $V = 12 \times 4/3 \pi \checkmark$.

On a sphere, heat conduction with uniform surface conditions depends only on the radius. The PDE reduces to a 1-D problem with standard series solutions, which can be mapped back, or used as a shape-factor approximation, for the egg.

Answer. $V = 16 \pi \approx 50.3 \text{ cm}^3$; $X = x/2$, $Y = y/2$, $Z = z/3$ gives the unit sphere.

7. Mechanical Engineering

Question 1: The cam profile

Question. A cam profile is given in polar form by $r = 40 + 10 \cos \theta$ (mm). Convert it to Cartesian form, find the maximum and minimum radius (the follower lift), and find the area enclosed by the profile.

Multiply $r = 40 + 10 \cos \theta$ by r to get $r^2 = 40r + 10r \cos \theta$, that is

$$x^2 + y^2 = 40 \sqrt{x^2 + y^2} + 10x$$

This curve is a limaçon.

$r_{\max} = 50$ mm at $\theta = 0$ and $r_{\min} = 30$ mm at $\theta = \pi$, so the follower lift is 20 mm.

$$A = \frac{1}{2} \int_0^{2\pi} (40 + 10 \cos \theta)^2 d\theta = \frac{1}{2} [1600(2\pi) + 0 + 100\pi] = 1650\pi \approx 5184 \text{ mm}^2$$

Answer. Lift 20 mm; area $1650\pi \approx 5184 \text{ mm}^2$.

Question 2: The two machine members

Question. Two machine members lie along the lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{6}$ and $\frac{x-2}{3} = \frac{y-1}{-2} = \frac{z+1}{6}$. Find the angle between them and the shortest distance between the lines.

The directions are $\mathbf{d}_1 = (2, 3, 6)$ and $\mathbf{d}_2 = (3, -2, 6)$, both of length 7:

$$\cos \alpha = \frac{6 - 6 + 36}{49} = \frac{36}{49} \Rightarrow \alpha = 42.7^\circ$$

For the shortest distance, take $P_1 = (1, -1, 0)$ and $P_2 = (2, 1, -1)$, so $\overrightarrow{P_1 P_2} = (1, 2, -1)$, and $\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = (30, 6, -13)$ with $\|\mathbf{n}\| = \sqrt{1105} = 33.24$:

$$d = \frac{|\overrightarrow{P_1 P_2} \cdot \mathbf{n}|}{\|\mathbf{n}\|} = \frac{|30 + 12 + 13|}{33.24} = 1.65$$

Answer. Angle 42.7° ; shortest distance 1.65 units.

8. Petroleum Engineering

Question 1: The drainage volume

Question. Radial flow into a vertical well is analysed in cylindrical coordinates. The drainage area is a circle of radius $r_e = 300$ m around a well of radius $r_w = 0.1$ m. Write the flow domain in cylindrical coordinates, and find the volume of a reservoir 20 m thick with 25 % porosity.

The flow domain is $0.1 \leq r \leq 300$ m, $0 \leq \theta < 2\pi$, $0 \leq z \leq 20$ m.

$$V_{\text{bulk}} = \pi (300^2 - 0.1^2) (20) = 5.65 \times 10^6 \text{ m}^3, V_{\text{pore}} = 0.25 V_{\text{bulk}} = 1.41 \times 10^6 \text{ m}^3$$

Answer. Bulk volume $5.65 \times 10^6 \text{ m}^3$; pore volume $1.41 \times 10^6 \text{ m}^3$.

Question 2: The directional well

Question. A directional well leaves the surface at $(0, 0, 0)$ and reaches a target at $(600, 800, -2500)$ m. Find the equation of the straight well path, its inclination from the vertical, and its azimuth. Express the target in spherical coordinates.

The path is $\mathbf{r}(s) = s(600, 800, -2500)$ for $0 \leq s \leq 1$, or $\frac{x}{600} = \frac{y}{800} = \frac{z}{-2500}$.

$$L = \sqrt{600^2 + 800^2 + 2500^2} = 2692.6 \text{ m}, \text{inclination} = \tan^{-1} \frac{1000}{2500} = 21.8^\circ$$

Taking y as north and x as east, the azimuth is $\tan^{-1}(600/800) = 36.9^\circ$ (N36.9°E). In spherical coordinates (ϕ measured from the positive z -axis):

$$\rho = 2692.6 \text{ m}, \theta = \tan^{-1} \frac{800}{600} = 53.1^\circ, \phi = \cos^{-1} \frac{-2500}{2692.6} = 158.2^\circ$$

Answer. Length 2692.6 m; inclination 21.8° ; azimuth N36.9°E;

$(\rho, \theta, \phi) = (2692.6 \text{ m}, 53.1^\circ, 158.2^\circ)$.

Topic 5 Solutions: Functions of Several Variables

1. Agricultural Engineering

Question 1: Maximum crop yield

Question. Crop yield (t/ha) is modelled by

$Y(N, W) = 0.04N + 0.05W - 0.0001N^2 - 0.0002W^2 + 0.00005NW$, where N is nitrogen (kg/ha) and W is water (mm). Find the input levels that maximise yield, and confirm that the point is a maximum.

Set both partial derivatives to zero:

$$Y_N = 0.04 - 0.0002N + 0.00005W = 0, Y_W = 0.05 - 0.0004W + 0.00005N = 0$$

Solving gives $N = 238.7$ kg/ha and $W = 154.8$ mm, with $Y_{max} = 8.65$ t/ha. For the second-derivative test, $Y_{NN} = -0.0002$, $Y_{WW} = -0.0004$ and $Y_{NW} = 0.00005$:

$$D = (-0.0002)(-0.0004) - (0.00005)^2 = 7.75 \times 10^{-8} > 0, Y_{NN} < 0$$

so the point is a **maximum**.

Answer. $N = 238.7$ kg/ha, $W = 154.8$ mm, $Y_{max} = 8.65$ t/ha.

Question 2: Error in evapotranspiration

Question. The evapotranspiration rate $E = kT^{0.5}R^{0.8}$ depends on temperature T and radiation R . Use the total differential to estimate the percentage error in E when T is measured with 2 % error and R with 3 % error.

Taking logarithms, $\ln E = \ln k + 0.5 \ln T + 0.8 \ln R$. Differentiating,

$$\frac{dE}{E} = 0.5 \frac{dT}{T} + 0.8 \frac{dR}{R}$$

In the worst case the errors add: $0.5(2\%) + 0.8(3\%) = 3.4\%$.

Answer. The maximum error in E is 3.4 %.

2. Chemical Engineering

Question 1: Van der Waals partial derivatives

Question. For a van der Waals gas, $\left(P + \frac{a}{V^2}\right)(V - b) = RT$. Find $\left(\frac{\partial P}{\partial V}\right)_T$ and $\left(\frac{\partial P}{\partial T}\right)_V$. Verify the cyclic relation $\left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial T}{\partial P}\right)_V = -1$.

Write $P = \frac{RT}{V - b} - \frac{a}{V^2}$. Then

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3}, \left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V-b}$$

Treating $F(P, V, T) = 0$ implicitly, $\left(\frac{\partial V}{\partial T}\right)_P = -\frac{(\partial P/\partial T)_V}{(\partial P/\partial V)_T}$ and $\left(\frac{\partial T}{\partial P}\right)_V = \frac{1}{(\partial P/\partial T)_V}$. Multiplying the three:

$$\left(\frac{\partial P}{\partial V}\right)_T \cdot \left(-\frac{(\partial P/\partial T)_V}{(\partial P/\partial V)_T}\right) \cdot \frac{1}{(\partial P/\partial T)_V} = -1 \blacksquare$$

Answer. The two partial derivatives are given above, and their cyclic product is -1 .

Question 2: Least surface area for a fixed volume

Question. A cylindrical reactor of fixed volume 10 m^3 is to be built with the least surface area (and so the least material cost). Use Lagrange multipliers to find the best radius and height.

Minimise $A = 2\pi r^2 + 2\pi r h$ subject to $\pi r^2 h = 10$. The Lagrange conditions are

$$4\pi r + 2\pi h = \lambda \cdot 2\pi r h, 2\pi r = \lambda \pi r^2$$

The second gives $\lambda = 2/r$; substituting into the first gives $4r + 2h = 4h$, so $h = 2r$ (the height equals the diameter). Then $2\pi r^3 = 10$, so

$$r = \left(\frac{5}{\pi}\right)^{1/3} = 1.168 \text{ m}, h = 2.335 \text{ m}, A_{\min} = 25.7 \text{ m}^2$$

Answer. $r = 1.168 \text{ m}$ and $h = 2.335 \text{ m}$ (height = diameter).

3. Computer Engineering

Question 1: Gradient descent

Question. Gradient descent minimises the loss $L(w_1, w_2) = (w_1 - 3)^2 + 2(w_2 + 1)^2 + w_1 w_2$. Compute ∇L , write the gradient-descent update rule, and carry out 3 iterations from $(0, 0)$ with learning rate 0.1 .

$$\nabla L = (2(w_1 - 3) + w_2, 4(w_2 + 1) + w_1), \mathbf{w} \leftarrow \mathbf{w} - \eta \nabla L(\mathbf{w}), \eta = 0.1$$

k	w_1	w_2	L
0	0	0	11.00
1	0.600	-0.400	6.24
2	1.120	-0.700	2.93
3	1.566	-0.932	0.61

The iterates are heading towards the true minimum, where $\nabla L = \mathbf{0}$: $(4, -2)$ with $L = -4$.

Answer. After 3 steps, $\mathbf{w} = (1.566, -0.932)$.

Question 2: Image gradient

Question. The brightness of a grayscale image is modelled as $I(x, y) = 100e^{-(x^2+y^2)/50}$. Compute the image gradient at $(3, 4)$, its magnitude (the edge strength) and its direction, and explain how edge-detection algorithms use this.

$$I(3, 4) = 100e^{-25/50} = 60.65, \text{ and } \nabla I = -\frac{2}{50}(x, y)I, \text{ so}$$

$$\nabla I(3, 4) = -0.04(3, 4)(60.65) = (-7.28, -9.70), \|\nabla I\| = 12.13$$

The gradient direction is $(-0.6, -0.8)$, pointing towards the bright centre. Edge detectors (Sobel, Canny) estimate ∇I with finite differences at each pixel and mark as edges the pixels where $\|\nabla I\|$ is a local maximum across the gradient direction and above a threshold. The edge itself runs perpendicular to ∇I .

Answer. $\nabla I = (-7.28, -9.70)$; edge strength 12.13; direction $(-0.6, -0.8)$.

4. Civil Engineering

Question 1: Slope and drainage at (10, 20)

Question. The ground surface of a site is $z = 50 - 0.01x^2 - 0.02y^2$ (m). At the point $(10, 20)$, find the slope in the direction of the unit vector $(3/5, 4/5)$, the direction of steepest descent (the drainage path), and the equation of the tangent plane.

$$z(10, 20) = 50 - 1 - 8 = 41 \text{ m and } \nabla z = (-0.02x, -0.04y) = (-0.2, -0.8).$$

$$D_{\mathbf{u}}z = \nabla z \cdot (3/5, 4/5) = -0.12 - 0.64 = -0.76$$

The ground falls 0.76 m per metre in that direction. Water drains along $-\nabla z = (0.2, 0.8)$, or $(0.243, 0.970)$ as a unit vector, where the slope is $\|\nabla z\| = 0.825$. The tangent plane is

$$z = 41 - 0.2(x - 10) - 0.8(y - 20) \Leftrightarrow 0.2x + 0.8y + z = 59$$

Answer. Slope -0.76 ; drainage along $(0.243, 0.970)$; tangent plane $0.2x + 0.8y + z = 59$.

Question 2: Strongest beam from a log

Question. A rectangular beam cut from a circular log of diameter d is strongest in bending when bh^2 is largest. Use Lagrange multipliers to find b and h subject to $b^2 + h^2 = d^2$.

Maximise $S = bh^2$ subject to $b^2 + h^2 = d^2$. The Lagrange conditions are $h^2 = 2\lambda b$ and $2bh = 2\lambda h$, so $\lambda = b$ and $h^2 = 2b^2$. Substituting into the constraint gives $3b^2 = d^2$:

$$b = \frac{d}{\sqrt{3}} \approx 0.577d, h = d\sqrt{\frac{2}{3}} \approx 0.816d$$

Answer. $b = d/\sqrt{3}$ and $h = d\sqrt{2/3}$ (so $h:b = \sqrt{2}:1$).

5. Electrical and Electronics Engineering

Question 1: Error in dissipated power

Question. The power dissipated in a resistor is $P = V^2/R$. Given $V = 220 \pm 2$ V and $R = 50 \pm 0.5 \Omega$, use partial derivatives and the total differential to estimate the maximum absolute and relative error in P .

$P = V^2/R = 48\,400/50 = 968$ W, and

$$dP = \frac{2V}{R} dV - \frac{V^2}{R^2} dR = 8.8 dV - 19.36 dR$$

In the worst case the errors add: $|dP| \leq 8.8(2) + 19.36(0.5) = 17.6 + 9.68 = 27.3$ W. The relative error is

$$\frac{|dP|}{P} \leq 2 \cdot \frac{2}{220} + \frac{0.5}{50} = 0.0182 + 0.0100 = 2.8\%$$

Answer. $P = 968 \pm 27.3$ W, a relative error of 2.8 %.

Question 2: The point-charge potential

Question. The electric potential near a point charge is $V(x, y, z) = \frac{k}{\sqrt{x^2 + y^2 + z^2}}$. Show that V satisfies Laplace's equation $\nabla^2 V = 0$ away from the origin, and find the equation of the equipotential surface through $(1, 2, 2)$.

Let $r^2 = x^2 + y^2 + z^2$, so $V = k/r$. Then $V_x = -kx/r^3$ and $V_{xx} = -k/r^3 + 3kx^2/r^5$, with V_{yy} and V_{zz} of the same form. Adding,

$$\nabla^2 V = -\frac{3k}{r^3} + \frac{3k(x^2 + y^2 + z^2)}{r^5} = -\frac{3k}{r^3} + \frac{3k}{r^3} = 0 \quad (r \neq 0)$$

At $(1, 2, 2)$, $r = 3$ and $V = k/3$.

Answer. $\nabla^2 V = 0$; the equipotential through $(1, 2, 2)$ is the sphere $x^2 + y^2 + z^2 = 9$.

6. Food Engineering

Question 1: Moisture during drying

Question. During drying, the moisture content is $M(t, T) = M_0 e^{-k(T)t}$ with $k(T) = 0.01 e^{0.05(T-60)}$. Find $\partial M / \partial t$ and $\partial M / \partial T$, and use the chain rule to find dM/dt when the dryer temperature rises as $T = 60 + 0.5t$.

$$\frac{\partial M}{\partial t} = -k(T)M, \quad \frac{\partial M}{\partial T} = -M_0 t k'(T) e^{-kt} = -0.05 k(T) t M$$

since $k' = 0.05k$. With $T = 60 + 0.5t$, $dT/dt = 0.5$, and the chain rule gives

$$\frac{dM}{dt} = \frac{\partial M}{\partial t} + \frac{\partial M}{\partial T} \frac{dT}{dt} = -k(T)M(1 + 0.025t), \quad k = 0.01 e^{0.025t}$$

The rising temperature speeds up drying in two ways: k grows, and the extra factor $(1+0.025t)$ appears.

Answer. $\frac{dM}{dt} = -0.01 e^{0.025t} M (1+0.025t)$

Question 2: Cheapest cereal box

Question. A rectangular box for packaging cereal must hold 3000 cm^3 . The top and bottom cost twice as much per cm^2 as the sides. Use Lagrange multipliers to find the dimensions that give the cheapest box.

Let the base be $a \times b$ and the height c . The cost is proportional to $C = 2(2ab) + 2ac + 2bc$, subject to $abc = 3000$. The Lagrange conditions give $a = b$ (by symmetry) and $c = 2a$. Then $2a^3 = 3000$:

$$a = b = 11.45 \text{ cm}, c = 22.89 \text{ cm}$$

The box is twice as tall as it is wide because its top and bottom cost more.

Answer. Base $11.45 \text{ cm} \times 11.45 \text{ cm}$; height 22.89 cm .

7. Mechanical Engineering

Question 1: Path of fastest heating

Question. The temperature distribution in a plate is $T(x, y) = 100 - x^2 - 2y^2$ ($^\circ\text{C}$). A heat-seeking sensor at $(2, 1)$ moves in the direction of fastest heating. Find that direction and the rate of temperature change in it, and find the path of the sensor.

$\nabla T = (-2x, -4y)$, so $\nabla T(2, 1) = (-4, -4)$. The sensor moves along $\frac{(-1, -1)}{\sqrt{2}}$, towards the hotter centre, at the rate $\|\nabla T\| = 4\sqrt{2} \approx 5.66$ $^\circ\text{C}$ per unit length.

The path follows the gradient: $\dot{x} = -2x$ and $\dot{y} = -4y$, so $x = 2e^{-2t}$ and $y = e^{-4t}$. Eliminating t :

$$y = \frac{x^2}{4}$$

Answer. Direction $(-1, -1)/\sqrt{2}$; rate $4\sqrt{2} \approx 5.66$ $^\circ\text{C}/\text{unit}$; path $y = x^2/4$ towards $(0, 0)$.

Question 2: The helicoid blade surface

Question. The surface of a turbine blade is given parametrically by $\mathbf{r}(u, v) = (u \cos v, u \sin v, 0.5v)$ for $1 \leq u \leq 2$ and $0 \leq v \leq \pi/2$ (a helicoid). Find $\mathbf{r}_u \times \mathbf{r}_v$, the unit normal and the element of surface area dS .

$$\mathbf{r}_u = (\cos v, \sin v, 0), \mathbf{r}_v = (-u \sin v, u \cos v, 1/2)$$

$$\mathbf{r}_u \times \mathbf{r}_v = (1/2 \sin v, -1/2 \cos v, u), \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{u^2 + 1/4}$$

Answer. $\mathbf{n} = \frac{(1/2 \sin v, -1/2 \cos v, u)}{\sqrt{u^2 + 1/4}}$ and $dS = \sqrt{u^2 + 1/4} du dv$.

8. Petroleum Engineering

Question 1: The pressure field around the well

Question. The pressure distribution around a producing well is $p(x, y) = p_e - \frac{q\mu}{4\pi kh} \ln \frac{x^2 + y^2}{r_e^2}$.

Find ∇p , and show that the flow (along $-\nabla p$) is directed radially towards the well.

Let $c = \frac{q\mu}{4\pi kh}$ and $r^2 = x^2 + y^2$. Then $p = p_e - c \ln(r^2/r_e^2)$, and

$$\nabla p = -\frac{2c(x, y)}{r^2} = -\frac{q\mu}{2\pi kh} \frac{\hat{\mathbf{r}}}{r}$$

This is purely radial, with magnitude falling off as $1/r$. So the flow along $-\nabla p$ is radial, and the Darcy

velocity is $u = \frac{k}{\mu} \cdot \frac{q\mu}{2\pi kh r} = \frac{q}{2\pi r h}$.

Note on the sign. With the formula exactly as written, p increases towards the well, so $-\nabla p$ points **outward**: that describes an injector. For a **producing** well the pressure must be lowest at the well, $p = p_e + c \ln(r^2/r_e^2)$. Then ∇p points outward and the flow $-\nabla p$ points radially **inward**, towards the well ✓. Students should spot and correct the sign.

Answer. ∇p is radial with magnitude $\frac{q\mu}{2\pi kh r}$; with the corrected sign the flow is radially into the well.

Question 2: Linearising B_o

Question. The oil formation volume factor $B_o(p, T)$ is approximated by $B_o = 1.2 + 0.0001(T - 150) - 0.00002(p - 3000)$. Find the linear (Taylor) approximation of B_o about $(p, T) = (3000, 150)$, and estimate B_o at (2800 psi, 160 °F).

B_o is already linear in p and T , so its first-order Taylor expansion about $(3000, 150)$ is the function itself. At (2800 psi, 160 °F):

$$B_o = 1.2 + 0.0001(10) - 0.00002(-200) = 1.2 + 0.001 + 0.004 = 1.205$$

Answer. $B_o = 1.205$ rb/stb

Topic 6 Solutions: Multiple Integrals and Vector Calculus

1. Agricultural Engineering

Question 1: Rainfall over the catchment

Question. The rainfall intensity over a rectangular catchment $0 \leq x \leq 2$ km, $0 \leq y \leq 3$ km is $I(x, y) = 10 + 2x + y$ (mm/h). Use a double integral to find the total volumetric rainfall rate (m^3/h) over the catchment and the average intensity.

$$\iint_R (10 + 2x + y) dA = \int_0^3 \int_0^2 (10 + 2x + y) dx dy = \int_0^3 (24 + 2y) dy = 72 + 9 = 81 \text{ mm} \cdot \text{km}^2/\text{h}$$

Since 1 mm is 10^{-3} m and 1 km^2 is 10^6 m^2 , this is $Q = 81\,000 \text{ m}^3/\text{h}$. The area is 6 km^2 , so the average intensity is $81/6 = 13.5 \text{ mm/h}$.

Answer. $Q = 81\,000 \text{ m}^3/\text{h}$; average intensity 13.5 mm/h .

Question 2: Soil water flux

Question. Water flows through a soil profile with Darcy velocity field $\mathbf{q} = (-2x, -2y, 4z) \times 10^{-6} \text{ m/s}$. Compute $\nabla \cdot \mathbf{q}$ and $\nabla \times \mathbf{q}$. State whether the flow is incompressible and irrotational, and find a potential function if one exists.

Ignoring the common factor 10^{-6} :

$$\nabla \cdot \mathbf{q} = -2 - 2 + 4 = 0, \nabla \times \mathbf{q} = \mathbf{0}$$

The curl is zero because each component depends only on its own variable. So the flow is **incompressible** and **irrotational**, and a potential exists with $\mathbf{q} = \nabla \phi$:

$$\phi = (-x^2 - y^2 + 2z^2) \times 10^{-6}$$

Check: $\nabla^2 \phi = 0$, so this is a valid potential flow.

Answer. Incompressible and irrotational, with $\phi = (-x^2 - y^2 + 2z^2) \times 10^{-6}$.

2. Chemical Engineering

Question 1: Catalyst mass

Question. A cylindrical packed-bed reactor has radius 0.5 m and length 2 m. The catalyst density varies as $\rho(r) = 800(1 - 0.3r^2) \text{ kg/m}^3$. Use a triple integral in cylindrical coordinates to find the total catalyst mass.

$$m = \int_0^2 \int_0^{2\pi} \int_0^{0.5} 800(1 - 0.3r^2) r dr d\theta dz = 3200\pi \left[\frac{r^2}{2} - 0.075r^4 \right]_0^{0.5} = 3200\pi(0.125 - 0.00469) = 1209.5 \text{ kg}$$

Answer. $m = 1209.5 \text{ kg}$

Question 2: Poiseuille flow rate

Question. The velocity in a pipe of radius R is $\mathbf{v} = v_{\max} \left(1 - \frac{r^2}{R^2}\right) \mathbf{k}$ (laminar Poiseuille flow).

Compute the volumetric flow rate as the flux of $\mathbf{v}$ through the pipe cross-section, and show that the mean velocity is $v_{\max}/2$.

$$Q = \iint \mathbf{v} \cdot \mathbf{k} \, dA = \int_0^{2\pi} \int_0^R v_{\max} \left(1 - \frac{r^2}{R^2}\right) r \, dr \, d\theta = 2\pi v_{\max} \left[\frac{R^2}{2} - \frac{R^2}{4} \right] = \frac{\pi R^2 v_{\max}}{2}$$

The mean velocity is $\dot{v} = Q/(\pi R^2) = v_{\max}/2$. ■

Answer. $Q = \pi R^2 v_{\max}/2$ and $\dot{v} = v_{\max}/2$.

3. Computer Engineering

Question 1: The drone's helical path

Question. A drone flies along $\mathbf{r}(t) = (10 \cos t, 10 \sin t, 2t)$ m for $0 \leq t \leq 2\pi$. Find its velocity, speed and acceleration, and the total path length. Write a short algorithm that approximates this arc length numerically with the trapezoidal rule, and compare the result with the exact value.

$$\mathbf{v} = (-10 \sin t, 10 \cos t, 2), \|\mathbf{v}\| = \sqrt{104} \approx 10.20 \text{ m/s}, \mathbf{a} = (-10 \cos t, -10 \sin t, 0)$$

The acceleration has magnitude 10 m/s^2 and points towards the axis. The path length is

$$s = \int_0^{2\pi} \sqrt{104} \, dt = 2\pi \sqrt{104} = 64.08 \text{ m}$$

```
function arc_length(r_prime, a, b, n):
    h = (b - a) / n
    s = 0.5 * (|r_prime(a)| + |r_prime(b)|)
    for i = 1..n-1: s += |r_prime(a + i*h)|
    return h * s
```

Because the speed is constant, the trapezoidal rule gives 64.08 m exactly for any n . For a non-uniform path the error falls as h^2 .

Answer. Speed $\sqrt{104} \approx 10.20 \text{ m/s}$; length $2\pi \sqrt{104} = 64.08 \text{ m}$.

Question 2: Heat generated in the heat sink

Question. A heat sink on a processor chip occupies the region $0 \leq x \leq 2, 0 \leq y \leq 2, 0 \leq z \leq 1$ (cm). The heat generation rate is $q(x, y, z) = 5xy + z$ (W/cm³). Use a triple integral to find the total heat generated, and write it as a nested-loop numerical approximation (a Riemann sum).

$$Q = \int_0^1 \int_0^2 \int_0^2 (5xy + z) \, dx \, dy \, dz = 5(2)(2)(1) + (2)(2)(1/2) = 20 + 2 = 22 \text{ W}$$

As a Riemann sum (midpoint rule):

```
Q = 0
for i in 0..n-1: for j in 0..n-1: for k in 0..n-1:
    x = (i+0.5)*dx; y = (j+0.5)*dy; z = (k+0.5)*dz
    Q += (5*x*y + z) * dx*dy*dz
```

The integrand is linear in each variable separately, so the midpoint rule gives exactly 22.

Answer. $Q = 22 \text{ W}$

4. Civil Engineering

Question 1: Centroid and second moment of the T-section

Question. A lamina shaped like a T-beam cross-section occupies the region between $y=0$ and $y=4$ for $-1 \leq x \leq 1$ (the web), plus the region $4 \leq y \leq 5$ for $-3 \leq x \leq 3$ (the flange). Use double integrals to find the centroid $\bar{y}$ and the second moment of area I_x about the centroidal axis.

The web has area 8 with centroid at $y=2$; the flange has area 6 with centroid at $y=4.5$.

$$\bar{y} = \frac{\iint y \, dA}{A} = \frac{8(2) + 6(4.5)}{14} = \frac{43}{14} = 3.071$$

$$I_x = \sum \left(\frac{bh^3}{12} + A d^2 \right) = \left[\frac{2 \cdot 4^3}{12} + 8(1.071)^2 \right] + \left[\frac{6 \cdot 1^3}{12} + 6(1.429)^2 \right] = \frac{1369}{42} \approx 32.6$$

Answer. $\bar{y} = 3.071$ above the base; $I_x \approx 32.6 \text{ units}^4$.

Question 2: Work done by the crane force

Question. A crane lifts a load along a path in the force field $\mathbf{F} = (2xy, x^2 + z, y) \text{ kN}$. Compute the work done $\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0, 0, 0)$ to $(1, 2, 3)$ along two different paths. Decide whether $\mathbf{F}$ is conservative, and find its potential if it is.

Path 1, the straight line $\mathbf{r} = t(1, 2, 3)$:

$$W = \int_0^1 [(2t)(2t)(1) + (t^2 + 3t)(2) + (2t)(3)] dt = \int_0^1 (6t^2 + 12t) dt = 2 + 6 = 8 \text{ kJ}$$

Path 2, the broken line $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 2, 0) \rightarrow (1, 2, 3)$: the first leg gives 0 (since $F_x = 0$ on $y=0$), the second $\int_0^2 1 \, dy = 2$, and the third $\int_0^3 2 \, dz = 6$. Total 8 kJ, the same as path 1.

$\nabla \times \mathbf{F} = (1 - 1, 0 - 0, 2x - 2x) = \mathbf{0}$, so $\mathbf{F}$ is conservative, with potential

$$\phi = x^2 y + yz, W = \phi(1, 2, 3) - \phi(0, 0, 0) = 2 + 6 = 8 \checkmark$$

Answer. $W = 8 \text{ kJ}$ on both paths; $\mathbf{F}$ is conservative with $\phi = x^2 y + yz$.

5. Electrical and Electronics Engineering

Question 1: Flux through the cylinder

Question. The electric field of an infinite line charge is $\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}}$. Compute the flux of $\mathbf{E}$ through a closed cylinder of radius a and length L coaxial with the line. Show that it equals $\lambda L / \epsilon_0$, as Gauss's law requires.

On the curved side, $\mathbf{E}$ is parallel to $\mathbf{n}$ with $\|\mathbf{E}\| = \frac{\lambda}{2\pi\epsilon_0 a}$ over an area $2\pi a L$. On the two flat ends, $\mathbf{E} \perp \mathbf{n}$, so the flux there is 0. The total flux is

$$\Phi = \frac{\lambda}{2\pi\epsilon_0 a} \cdot 2\pi a L = \frac{\lambda L}{\epsilon_0}$$

This is the enclosed charge divided by ϵ_0 , as Gauss's law requires. ■

Answer. $\Phi = \lambda L / \epsilon_0$

Question 2: Circulation of B

Question. The magnetic field around a long straight wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi r} \hat{\boldsymbol{\theta}}$. Evaluate the line integral $\oint \mathbf{B} \cdot d\mathbf{r}$ around a circle of radius a centred on the wire, and compute $\nabla \times \mathbf{B}$ for $r \neq 0$. Comment on the results.

On $r=a$, $\mathbf{B} \cdot d\mathbf{r} = \frac{\mu_0 I}{2\pi a} a d\theta$, so

$$\oint \mathbf{B} \cdot d\mathbf{r} = \int_0^{2\pi} \frac{\mu_0 I}{2\pi} d\theta = \mu_0 I, \quad \nabla \times \mathbf{B} = \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{\mu_0 I}{2\pi} \right) \mathbf{k} = \mathbf{0} \quad (r \neq 0)$$

The field is irrotational everywhere except on the wire, yet the circulation around the wire is not zero. The two results are consistent: the region without the wire is **not simply connected**, and all of the curl is concentrated on the wire ($\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$).

Answer. $\oint \mathbf{B} \cdot d\mathbf{r} = \mu_0 I$ (Ampère's law), and $\nabla \times \mathbf{B} = \mathbf{0}$ off the wire.

6. Food Engineering

Question 1: Average temperature in the dome cake

Question. A hemispherical dome-shaped cake of radius 10 cm has a temperature during baking of $T(\rho) = 180 - 60 \left(1 - \frac{\rho}{10} \right)^\circ \text{C}$. Use a triple integral in spherical coordinates to find the volume-averaged temperature.

$T = 180 - 60(1 - \rho/10) = 120 + 6\rho$. The angular integrals cancel between numerator and denominator, leaving

$$\bar{T} = \frac{\int_0^{10} (120 + 6\rho) \rho^2 d\rho}{\int_0^{10} \rho^2 d\rho} = \frac{40000 + 15000}{1000/3} = 165^\circ \text{C}$$

Answer. $\bar{T} = 165^\circ \text{C}$

Question 2: Air and moisture flow through the dryer opening

Question. Air flows through a rectangular dryer opening $0 \leq x \leq 1$ m, $0 \leq z \leq 0.5$ m with velocity field $\mathbf{v} = (0, 2+x-z, 0)$ m/s. Compute the volumetric air flow rate (the flux through the opening). If the air carries 0.012 kg of water per m^3 , find the rate of moisture removal.

The opening lies in a plane $y = \text{const}$ with normal $\mathbf{j}$, so only v_y counts:

$$Q = \int_0^1 \int_0^{0.5} (2+x-z) dz dx = \int_0^1 (0.875 + 0.5x) dx = 1.125 \text{ m}^3/\text{s}$$

The moisture removal rate is $1.125 \times 0.012 = 0.0135$ kg/s ≈ 48.6 kg/h.

Answer. $Q = 1.125 \text{ m}^3/\text{s}$; moisture removal 0.0135 kg/s (48.6 kg/h).

7. Mechanical Engineering

Question 1: Moment of inertia of a solid cone

Question. Find the mass moment of inertia about the z -axis of a solid cone of height h , base radius a and uniform density ρ , using a triple integral in cylindrical coordinates.

Put the apex at the origin, so the radius at height z is az/h :

$$I_z = \rho \int_0^h \int_0^{2\pi} \int_0^{az/h} r^2 \cdot r dr d\theta dz = 2\pi \rho \int_0^h \frac{a^4 z^4}{4h^4} dz = \frac{\pi \rho a^4 h}{10}$$

With $m = \rho \pi a^2 h/3$, this is

$$I_z = \frac{3}{10} m a^2$$

Answer. $I_z = 3/10 m a^2$

Question 2: Rigid-body rotation

Question. A fluid velocity field is $\mathbf{v} = (y, -x, 0)$. Compute $\nabla \cdot \mathbf{v}$ and $\nabla \times \mathbf{v}$, interpret the flow physically (rigid-body rotation), and evaluate the circulation around the unit circle in the xy -plane.

$$\nabla \cdot \mathbf{v} = 0 + 0 = 0, \nabla \times \mathbf{v} = \left(0, 0, \frac{\partial(-x)}{\partial x} - \frac{\partial y}{\partial y} \right) = (0, 0, -2)$$

This is **rigid-body rotation**, clockwise seen from above, with angular velocity $\omega = 1/2 \|\nabla \times \mathbf{v}\| = 1$ rad/s.

Around the unit circle, taken counter-clockwise with $\mathbf{r} = (\cos t, \sin t)$:

$$\oint \mathbf{v} \cdot d\mathbf{r} = \int_0^{2\pi} [(\sin t)(-\sin t) + (-\cos t)(\cos t)] dt = -2\pi$$

This agrees with Stokes' theorem: $(-2)(\pi \cdot 1^2) = -2\pi$.

Answer. $\nabla \cdot \mathbf{v} = 0$, $\nabla \times \mathbf{v} = (0, 0, -2)$; circulation -2π .

8. Petroleum Engineering

Question 1: Bulk volume and oil in place

Question. A reservoir occupies the region between the surfaces $z = -2000$ and $z = -2000 - 0.001(x^2 + y^2)$ for $x^2 + y^2 \leq 500^2$ (m). Use a triple integral in cylindrical coordinates to find the bulk rock volume. With 20 % porosity and 75 % oil saturation, estimate the oil in place.

The thickness at radius r is $0.001 r^2$, so

$$V_{\text{bulk}} = \int_0^{2\pi} \int_0^{500} \int_0^{0.001 r^2} r \, dz \, dr \, d\theta = 0.001 \cdot 2\pi \cdot \frac{500^4}{4} = 9.82 \times 10^7 \text{ m}^3$$

$$\text{OIP} = \phi S_o V_{\text{bulk}} = 0.2 \times 0.75 \times 9.82 \times 10^7 = 1.47 \times 10^7 \text{ m}^3$$

Answer. Bulk volume $9.82 \times 10^7 \text{ m}^3$; oil in place $1.47 \times 10^7 \text{ m}^3$ (about 92.6 million barrels).

Question 2: Darcy flux across the square

Question. The pressure field in a reservoir is $p(x, y) = 3000 - 0.5x^2 - 0.3y^2$ (psi), and the Darcy velocity is $\mathbf{u} = -\frac{k}{\mu} \nabla p$. Find $\mathbf{u}$, and compute the flux of $\mathbf{u}$ across the boundary of the square $-100 \leq x, y \leq 100$ m (per unit thickness) with $k/\mu = 0.01$.

$\nabla p = (-x, -0.6y)$, so $\mathbf{u} = -0.01 \nabla p = (0.01x, 0.006y)$. On each side of the square (length 200 m):

- $x = \pm 100$: $\mathbf{u} \cdot \mathbf{n} = 1$, giving $1 \times 200 = 200$ per side.
- $y = \pm 100$: $\mathbf{u} \cdot \mathbf{n} = 0.6$, giving $0.6 \times 200 = 120$ per side.

The net outward flux is $2(200) + 2(120) = 640$. Check with the divergence theorem: $\nabla \cdot \mathbf{u} = 0.016$, and $0.016 \times 200^2 = 640$ ✓. The positive value means this pressure field implies a net source, such as injection, inside the square.

Answer. $\mathbf{u} = (0.01x, 0.006y)$; net outward flux 640 (m^2 per unit time, per unit thickness).

Topic 7 Solutions: Integral Theorems and Multivalued Functions

1. Agricultural Engineering

Question 1: Field area by Green's theorem

Question. A field boundary is traced by GPS as the ellipse $x = 300 \cos t$, $y = 200 \sin t$ (m). Use Green's theorem ($A = 1/2 \oint (x dy - y dx)$) to find the field area. Explain how a GPS area-measuring device can use the same formula on a list of boundary points (the shoelace formula).

With $x = 300 \cos t$ and $y = 200 \sin t$, we have $dx = -300 \sin t dt$ and $dy = 200 \cos t dt$, so

$$A = \frac{1}{2} \oint (x dy - y dx) = \frac{1}{2} \int_0^{2\pi} (60\,000 \cos^2 t + 60\,000 \sin^2 t) dt = \frac{1}{2} (60\,000) (2\pi) = 60\,000\pi$$

For GPS points $(x_1, y_1), \dots, (x_n, y_n)$, treat the boundary as straight segments. The same line integral then becomes the **shoelace formula**, which is what handheld area meters compute:

$$A = \frac{1}{2} \left| \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i) \right|$$

Answer. $A = 60\,000\pi \approx 188\,496 \text{ m}^2$ (18.85 ha).

Question 2: Checking the divergence theorem

Question. The groundwater flux under a farm is $\mathbf{q} = (x^2, y^2, z^2)$ (m/day, scaled) in the cube $0 \leq x, y, z \leq 1$. Check the divergence theorem by computing the net outward flux both as a surface integral and as a volume integral.

Volume integral. $\nabla \cdot \mathbf{q} = 2x + 2y + 2z$, so

$$\iiint_V (2x + 2y + 2z) dV = 1 + 1 + 1 = 3$$

Surface integral. On the face $x = 1$, $\mathbf{q} \cdot \mathbf{n} = 1^2 = 1$, which integrates to 1. On $x = 0$, $\mathbf{q} \cdot \mathbf{n} = -0^2 = 0$. The y and z faces work the same way, so the total is $1 + 1 + 1 = 3$ ✓.

Answer. Both integrals equal 3.

2. Chemical Engineering

Question 1: Deriving the continuity equation

Question. Apply Gauss's divergence theorem to a fixed control volume V to derive the continuity equation $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$ from the integral mass balance.

The mass balance on a fixed control volume V says the rate of increase of mass equals the net inflow:

$$\frac{d}{dt} \iiint_V \rho dV = - \oint_S \rho \mathbf{v} \cdot \mathbf{n} dS$$

By the divergence theorem, $\oint_S \rho \mathbf{v} \cdot \mathbf{n} dS = \iiint_V \nabla \cdot (\rho \mathbf{v}) dV$. Since V is fixed,

$$\iiint_V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] dV = 0$$

This holds for every V , so the integrand must be zero everywhere. ■

Answer. $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$

Question 2: Values of $z^{2/3}$ at $8i$

Question. The complex potential of flow in a corner-shaped mixing channel involves $w = z^{2/3}$. Find all values of w at $z = 8i$, give the principal value, and state the branch point. Suggest a suitable branch cut for a channel occupying the upper half-plane.

$z = 8i = 8e^{i(\pi/2 + 2k\pi)}$, so

$$w = 8^{2/3} e^{i \frac{2}{3} \left(\frac{\pi}{2} + 2k\pi \right)} = 4 e^{i \left(\frac{\pi}{3} + \frac{4k\pi}{3} \right)}$$

- $k=0$: $4e^{i\pi/3} = 2 + 2\sqrt{3}i$ (the principal value)
- $k=1$: $4e^{i5\pi/3} = 2 - 2\sqrt{3}i$
- $k=2$: $4e^{i3\pi} = -4$

The branch point is $z=0$ (and $z=\infty$). For a channel in the upper half-plane, place the branch cut along the **negative imaginary axis**, taking $-\pi/2 < \arg z < 3\pi/2$. The whole flow region then has no cut in it, and w is single-valued and continuous there.

Answer. $w = 2 + 2\sqrt{3}i$ (principal), $2 - 2\sqrt{3}i$ and -4 ; branch point $z=0$; cut along the negative imaginary axis.

3. Computer Engineering

Question 1: The shoelace formula

Question. Polygon-area routines in computer graphics are based on Green's theorem. Derive the shoelace formula $A = 1/2 \sum_i (x_i y_{i+1} - x_{i+1} y_i)$ from $1/2 \oint (x dy - y dx)$, and use it on the polygon with vertices $(0,0)$, $(4,0)$, $(4,3)$, $(2,5)$ and $(0,3)$. Write the algorithm in pseudocode.

On the straight edge from (x_i, y_i) to (x_{i+1}, y_{i+1}) , parametrise $x = x_i + t \Delta x$, $y = y_i + t \Delta y$ for $0 \leq t \leq 1$. Then

$$\int (x dy - y dx) = \int_0^1 [(x_i + t \Delta x) \Delta y - (y_i + t \Delta y) \Delta x] dt = x_i \Delta y - y_i \Delta x = x_i y_{i+1} - x_{i+1} y_i$$

Summing over the edges gives $A = 1/2 \sum_i (x_i y_{i+1} - x_{i+1} y_i)$. ■ For the given vertices the terms are 0, 12, 14, 6 and 0, so the sum is 32 and $A = 16$.

```
function polygon_area(xs, ys):
    n = len(xs); s = 0
    for i in 0..n-1:
        j = (i + 1) mod n
        s += xs[i]*ys[j] - xs[j]*ys[i]
    return abs(s) / 2
```

Answer. $A = 16$ square units.

Question 2: atan2 and phase unwrapping

Question. A function $\text{atan2}(y, x)$ returns the principal argument of $z = x + i y$, which satisfies $-\pi < \theta \leq \pi$. Explain why $\arg z$ is multivalued. Show what happens to the computed angle as a point travels once round the origin, and describe how “phase unwrapping” algorithms fix the resulting jump.

$\arg z = \theta + 2k\pi$ for any integer k , so it has infinitely many values; atan2 always returns the principal value, with $-\pi < \theta \leq \pi$. As a point goes once round the origin counter-clockwise, the true angle rises steadily from 0 to 2π . atan2 , however, rises to π and then **jumps to** $-\pi$ when the point crosses the negative real axis, which is the branch cut.

Phase unwrapping removes these jumps: if two consecutive samples differ by more than π , add or subtract 2π from every later sample.

```
for i = 1..n-1:
    d = phase[i] - phase[i-1]
    if d > pi: offset -= 2pi
    if d < -pi: offset += 2pi
    unwrapped[i] = phase[i] + offset
```

Answer. atan2 jumps by 2π at the negative real axis; unwrapping adds back multiples of 2π .

4. Civil Engineering

Question 1: Area and centroid of the slab section

Question. Use Green’s theorem to find the area and centroid of the plane region bounded by $y = x^2$ and $y = 2x$ (in metres), which models a curved slab section. Check the answer with direct double integration.

The curves meet at $x = 0$ and $x = 2$. Going round counter-clockwise, along $y = x^2$ from $(0, 0)$ to $(2, 4)$ and back along $y = 2x$:

$$A = \frac{1}{2} \oint (x dy - y dx) = \frac{1}{2} \left[\int_0^2 (2x^2 - x^2) dx + \int_2^0 (2x - 2x) dx \right] = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}$$

$$\dot{x} = \frac{1}{A} \oint \frac{x^2}{2} dy = 1, \quad \dot{y} = -\frac{1}{A} \oint \frac{y^2}{2} dx = \frac{8}{5}$$

Direct check: $A = \int_0^2 (2x - x^2) dx = 4 - 8/3 = 4/3 \checkmark$ and $\dot{x} = \frac{1}{A} \int_0^2 x(2x - x^2) dx = 1 \checkmark$.

Answer. $A = 4/3 \text{ m}^2$; centroid $(1, 1.6) \text{ m}$.

Question 2: Seepage under the dam

Question. Seepage under a dam has flux field $\mathbf{q} = (2x - y, x + 3y, -5z)$ (per unit time). Check whether the field is consistent with an incompressible soil ($\nabla \cdot \mathbf{q} = 0$). Use the divergence theorem to find the net outflow from the unit cube.

$$\nabla \cdot \mathbf{q} = 2 + 3 - 5 = 0$$

The field is consistent with an incompressible soil and no sources or sinks. By the divergence theorem, the net outflow from the unit cube is $\iiint 0 \, dV = 0$: whatever flows in, flows out.

Answer. $\nabla \cdot \mathbf{q} = 0$; the net outflow is zero.

5. Electrical and Electronics Engineering

Question 1: Ampère's law in differential form

Question. Starting from Ampère's circuital law $\oint_C \mathbf{H} \cdot d\mathbf{r} = I_{\text{enc}}$, use Stokes' theorem to derive the differential form $\nabla \times \mathbf{H} = \mathbf{J}$. Apply it to find $\mathbf{H}$ inside a long straight conductor of radius a carrying a uniformly distributed current I .

$\oint_C \mathbf{H} \cdot d\mathbf{r} = I_{\text{enc}} = \iint_S \mathbf{J} \cdot \mathbf{n} \, dS$, and by Stokes' theorem the left side equals $\iint_S (\nabla \times \mathbf{H}) \cdot \mathbf{n} \, dS$. So

$$\iint_S (\nabla \times \mathbf{H} - \mathbf{J}) \cdot \mathbf{n} \, dS = 0$$

for every surface S , hence $\nabla \times \mathbf{H} = \mathbf{J}$. ■

Inside the conductor ($r < a$) the current density is uniform, so a circle of radius r encloses $I r^2 / a^2$. Then $H \cdot 2\pi r = I r^2 / a^2$, giving

$$H = \frac{I r}{2\pi a^2} \hat{\boldsymbol{\theta}}$$

which rises linearly to $I / (2\pi a)$ at the surface.

Answer. $\nabla \times \mathbf{H} = \mathbf{J}$; inside the conductor $H = I r / (2\pi a^2)$.

Question 2: $\log(-1 + i\sqrt{3})$

Question. The phase of an AC signal is found from the complex logarithm of the phasor. Find all values of $\log(-1 + i\sqrt{3})$, give the principal value, and explain what the choice of branch means when measuring phase angles above 180° .

$|z| = 2$ and $\arg z = 2\pi/3 + 2k\pi$, so

$$\log z = \ln 2 + i \left(\frac{2\pi}{3} + 2k\pi \right), k = 0, \pm 1, \pm 2, \dots$$

The principal value is $\log z = 0.693 + 2.094i$.

The principal branch gives phase angles only in the range $-180^\circ < \theta \leq 180^\circ$, so a phase of 240° is reported as -120° . When a phase is tracked continuously, for example in a PLL or when measuring the phase shift through a filter, you must stay on one branch and unwrap, not rely on the principal value.

Answer. $\log z = \ln 2 + i(2\pi/3 + 2k\pi)$; principal value $0.693 + 2.094i$.

6. Food Engineering

Question 1: Heat balance for the can

Question. A food can is heated in a retort. Heat flux into the can is $\mathbf{q} = -k \nabla T$. Use the divergence theorem on the can volume to show that the total rate of heat entering through the surface equals $\iiint_V \rho c_p \frac{\partial T}{\partial t} dV$ (when there is no internal generation). This is the basis of the heat-conduction equation.

The energy balance on the can volume V with surface S is

$$\iiint_V \rho c_p \frac{\partial T}{\partial t} dV = - \oiint_S \mathbf{q} \cdot \mathbf{n} dS = \oiint_S k \nabla T \cdot \mathbf{n} dS = \iiint_V \nabla \cdot (k \nabla T) dV$$

using the divergence theorem in the last step. So $\iiint_V [\rho c_p \partial T / \partial t - \nabla \cdot (k \nabla T)] dV = 0$ for any sub-volume, which gives $\rho c_p \partial T / \partial t = \nabla \cdot (k \nabla T)$. ■

Answer. For constant k this is the heat equation $\frac{\partial T}{\partial t} = \alpha \nabla^2 T$.

Question 2: Circulation round the bowl rim

Question. The stirring flow in a mixing bowl is approximated by $\mathbf{v} = (-y, x, 0)$ (rad/s units). Use Stokes' theorem to compute the circulation around the bowl rim (a circle of radius 0.15 m in the plane $z = 0.1$ m) through the flat disc it bounds. Check the answer against a direct line integral.

$\nabla \times \mathbf{v} = (0, 0, 2)$. On the flat disc with $\mathbf{n} = \mathbf{k}$:

$$\iint (\nabla \times \mathbf{v}) \cdot \mathbf{k} dS = 2 \times \pi (0.15)^2 = 0.045\pi \approx 0.141 \text{ m}^2/\text{s}$$

Direct check: with $\mathbf{r} = (0.15 \cos t, 0.15 \sin t, 0.1)$, $\mathbf{v} \cdot d\mathbf{r} = (0.15)^2 (\sin^2 t + \cos^2 t) dt$, so $\oint = 0.0225 \times 2\pi = 0.045\pi$ ✓.

Answer. Circulation $0.045\pi \approx 0.141 \text{ m}^2/\text{s}$.

7. Mechanical Engineering

Question 1: Work round the triangle

Question. Compute the work done by the force $\mathbf{F} = (y+z, z+x, x+y)$ round the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ using Stokes' theorem. Explain the result in terms of conservative forces.

$$\nabla \times \mathbf{F} = \left(\frac{\partial(x+y)}{\partial y} - \frac{\partial(z+x)}{\partial z}, \frac{\partial(y+z)}{\partial z} - \frac{\partial(x+y)}{\partial x}, \frac{\partial(z+x)}{\partial x} - \frac{\partial(y+z)}{\partial y} \right) = (0, 0, 0)$$

By Stokes' theorem, $W = \iint (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = 0$. $\mathbf{F}$ is conservative, with $\phi = x y + y z + z x$: the work round any closed path is zero, and the work between two points does not depend on the path.

Answer. $W=0$; $\mathbf{F} = \nabla(x y + y z + z x)$ is conservative.

Question 2: Flow round the re-entrant corner

Question. The flow round a sharp re-entrant corner of angle $3\pi/2$ in a duct is modelled by $w = z^{2/3}$. Identify the branch point, describe the branch cut, and find the two other values of w at $z=1$ besides the principal value. Explain why the velocity becomes unbounded at the corner.

The branch point is $z=0$, the corner (the other is at infinity). Place the branch cut along a ray inside the solid wall, outside the $3\pi/2$ flow sector, so the flow domain contains no cut.

At $z=1 = e^{i2k\pi}$, the values are $w = e^{i4k\pi/3}$:

- $k=0$: 1 (principal)
- $k=1$: $e^{i4\pi/3} = -1/2 - \sqrt{3}/2i$
- $k=2$: $e^{i8\pi/3} = -1/2 + \sqrt{3}/2i$

The velocity is

$$\frac{dw}{dz} = \frac{2}{3} z^{-1/3}, \quad \left| \frac{dw}{dz} \right| = \frac{2}{3} |z|^{-1/3} \rightarrow \infty \text{ as } z \rightarrow 0$$

The flow cannot turn a sharp convex corner smoothly, so the velocity becomes infinite there. In a real fluid, viscosity produces **flow separation** at such corners.

Answer. Branch point $z=0$; other values $-1/2 \mp \sqrt{3}/2i$; $|dw/dz| \propto |z|^{-1/3}$ is unbounded at the corner.

8. Petroleum Engineering

Question 1: Flux with and without a well

Question. Use the divergence theorem to show that for steady incompressible flow in a reservoir with no wells ($\nabla \cdot \mathbf{u} = 0$), the net volumetric flux through any closed surface is zero. Extend the argument to a region containing a producing well of rate q , and derive the resulting flux condition.

With no wells, $\nabla \cdot \mathbf{u} = 0$ everywhere in V , so by the divergence theorem

$$\oint_S \mathbf{u} \cdot \mathbf{n} dS = \iiint_V \nabla \cdot \mathbf{u} dV = 0$$

With a producing well of rate q inside V , the well is a point (or line) sink, $\nabla \cdot \mathbf{u} = -q \delta(\mathbf{r} - \mathbf{r}_w)$, and

$$\oint_S \mathbf{u} \cdot \mathbf{n} dS = -q$$

The net *inflow* through any closed surface around the well equals the production rate q , whatever the shape of the surface. This is the basis of the well boundary condition $2\pi r h u_r = -q$ at $r=r_w$.

Answer. Zero net flux without wells; $\oint \mathbf{u} \cdot \mathbf{n} \, dS = -q$ around a producer.

Question 2: The multivalued stream function

Question. The complex potential for flow towards a well at the origin is $w = \frac{q}{2\pi} \log z$. Show that the stream function $\psi = \text{Im } w$ is multivalued, find its change on going once round the well, and relate this jump to the well's production rate.

$w = \frac{q}{2\pi} (\ln r + i\theta)$, so $\psi = \text{Im } w = \frac{q}{2\pi} \theta$. Since θ increases by 2π on each circuit, ψ increases by q each time round the well: ψ is multivalued, with its branch point at the well.

The difference in ψ between two streamlines equals the volumetric flow between them. A jump of exactly q on going once round means the total flow crossing any closed curve around the well equals the well's production rate q (per unit thickness).

Answer. $\psi = q\theta/(2\pi)$ jumps by q per circuit, which is the production rate.

Topic 8 Solutions: Complex Analysis

1. Agricultural Engineering

Question 1: Groundwater flow under the drainage field

Question. Steady groundwater flow under a drainage field is modelled by the potential $\phi(x, y) = x^2 - y^2$. Check that ϕ is harmonic. Use the Cauchy-Riemann equations to find the stream function ψ and the complex potential $w = \phi + i\psi$, and sketch the flow lines. Where is the stagnation point?

$\phi_{xx} + \phi_{yy} = 2 - 2 = 0$, so ϕ is harmonic. From the Cauchy-Riemann equations:

$$\psi_y = \phi_x = 2x \Rightarrow \psi = 2xy + g(x), \quad \psi_x = -\phi_y = 2y \Rightarrow g'(x) = 0$$

So $\psi = 2xy$ and

$$w = \phi + i\psi = (x + iy)^2 = z^2$$

The streamlines $xy = \text{const}$ are rectangular hyperbolae: this is flow into a corner. The complex velocity $dw/dz = 2z$ vanishes only at $z = 0$.

Answer. $\psi = 2xy$, $w = z^2$; the stagnation point is the origin.

Question 2: Impulse response of the greenhouse controller

Question. The transfer function of a greenhouse temperature controller is

$$G(s) = \frac{10}{(s+1)(s+2)(s+5)}. \text{ Find its poles and zeros, and use the residue theorem to find the impulse response } g(t), \text{ the inverse Laplace transform, as } \frac{1}{2\pi i} \oint e^{st} G(s) ds.$$

G has no finite zeros and simple poles at $s = -1, -2, -5$. The residues of $e^{st} G(s)$ are

$$\text{Res}_{s=-1} = \frac{10e^{-t}}{(1)(4)} = 2.5e^{-t}, \quad \text{Res}_{s=-2} = \frac{10e^{-2t}}{(-1)(3)} = -\frac{10}{3}e^{-2t}, \quad \text{Res}_{s=-5} = \frac{10e^{-5t}}{(-4)(-3)} = \frac{5}{6}e^{-5t}$$

Check: $g(0) = 2.5 - 3.333 + 0.833 = 0 \checkmark$, as expected for a third-order system.

Answer. $g(t) = 2.5e^{-t} - 10/3 e^{-2t} + 5/6 e^{-5t}$ for $t \geq 0$.

2. Chemical Engineering

Question 1: Which functions are analytic

Question. Use the Cauchy-Riemann equations to show that $f(z) = \dot{z} = x - iy$ is not analytic anywhere, while $f(z) = e^x(\cos y + i \sin y)$ is analytic everywhere. Identify the second function as e^z . Explain why analyticity is needed when the concentration field in a thin 2-D film is modelled as the real part of an analytic function.

For $f = \dot{z}$: $u = x$ and $v = -y$, so $u_x = 1$ but $v_y = -1$. The first Cauchy-Riemann equation fails at every point, so $\dot{z}$ is analytic nowhere.

For $f = e^x(\cos y + i \sin y)$: $u = e^x \cos y$ and $v = e^x \sin y$, so

$$u_x = e^x \cos y = v_y, u_y = -e^x \sin y = -v_x$$

The partial derivatives are continuous, so f is analytic everywhere, and $e^{x+iy} = e^x e^{iy}$ shows that $f = e^z$.

If the concentration is $c = \operatorname{Re} f(z)$ with f analytic, then c automatically satisfies Laplace's equation (steady diffusion without reaction), and the iso-concentration lines and flux lines ($\operatorname{Im} f$) are orthogonal. Without analyticity none of this holds.

Answer. $\dot{z}$ is nowhere analytic; $e^x(\cos y + i \sin y) = e^z$ is analytic everywhere.

Question 2: Step response of the process

Question. The response of a first-order-plus-lag process is $C(s) = \frac{1}{s(\tau s + 1)^2}$. Classify its singularities, and use residues to find $c(t)$ for $\tau = 2$ min.

$C(s) = \frac{1}{s(2s+1)^2} = \frac{1/4}{s(s+1/2)^2}$ has a simple pole at $s=0$, a double pole at $s=-1/2$, and no zeros.

$$\operatorname{Res}_{s=0} = \frac{1}{4(1/2)^2} = 1, \quad \operatorname{Res}_{s=-1/2} = \frac{d}{ds} \left[\frac{e^{st}}{4s} \right]_{s=-1/2} = -\frac{t}{2} e^{-t/2} - e^{-t/2}$$

Answer. $c(t) = 1 - (1+t/2)e^{-t/2}$ (t in minutes).

3. Computer Engineering

Question 1: Filter stability

Question. A digital filter has transfer function $H(z) = \frac{z^2 - 1}{z^2 - 0.5z + 0.06}$. Find its zeros and poles, and state whether the filter is stable (all poles inside the unit circle). Write a program sketch that computes the poles and tests stability for any given denominator coefficients.

The zeros are the roots of $z^2 - 1 = 0$, so $z = \pm 1$. The poles are the roots of $z^2 - 0.5z + 0.06 = 0$, so $z = 0.2$ and $z = 0.3$. Both poles lie inside the unit circle, so the filter is **stable**. The zeros on the unit circle block DC ($z = 1$) and the Nyquist frequency ($z = -1$), so this is a band-pass filter.

```
import numpy as np
def is_stable(den): # den: denominator coefficients
    poles = np.roots(den)
    return all(abs(p) < 1 for p in poles), poles
```

Answer. Zeros ± 1 ; poles 0.2 and 0.3; stable.

Question 2: Inverse z-transform

Question. The inverse z-transform is $x[n] = \frac{1}{2\pi i} \oint X(z) z^{n-1} dz$. Use the residue theorem to find $x[n]$ for $X(z) = \frac{z}{(z-0.5)(z-0.25)}$ in the region outside the circle of radius 0.5.

$X(z) z^{n-1} = \frac{z^n}{(z-1/2)(z-1/4)}$. For $n \geq 0$ both poles lie inside the contour:

$$\text{Res}_{z=1/2} = \frac{(1/2)^n}{1/2 - 1/4} = 4(1/2)^n, \text{Res}_{z=1/4} = \frac{(1/4)^n}{1/4 - 1/2} = -4(1/4)^n$$

Answer. $x[n] = 4[(1/2)^n - (1/4)^n]$ for $n \geq 0$, that is 0, 1, 0.75, 0.4375, ...

4. Civil Engineering

Question 1: Seepage under the sheet pile

Question. Seepage under a sheet pile is modelled by the complex potential $w = \cosh^{-1}(z/b)$. Rewrite it as $z = b \cosh w$, separate the real and imaginary parts, and show that the equipotential lines are ellipses and the streamlines are hyperbolae. Identify the branch points at $z = \pm b$.

Let $z = b \cosh w$ with $w = \phi + i\psi$. Then

$$x = b \cosh \phi \cos \psi, y = b \sinh \phi \sin \psi$$

Holding ϕ fixed and eliminating ψ gives the equipotentials, which are confocal ellipses:

$$\frac{x^2}{b^2 \cosh^2 \phi} + \frac{y^2}{b^2 \sinh^2 \phi} = 1$$

Holding ψ fixed and eliminating ϕ gives the streamlines, which are confocal hyperbolae:

$$\frac{x^2}{b^2 \cos^2 \psi} - \frac{y^2}{b^2 \sin^2 \psi} = 1$$

Both families have their foci at $z = \pm b$, which are the branch points of $\cosh^{-1}(z/b)$. There $dz/dw = b \sinh w = 0$, so the inverse map is not analytic.

Answer. Equipotentials are ellipses and streamlines are hyperbolae, all with foci at the branch points $z = \pm b$.

Question 2: $\int_{-\infty}^{\infty} dx / (x^2 + 4)^2$

Question. Evaluate the real integral $\int_{-\infty}^{\infty} \frac{dx}{(x^2 + 4)^2}$ by contour integration over a semicircle in the upper half-plane. The integral appears in the deflection of a beam on an elastic foundation under a distributed load.

$f(z) = \frac{1}{(z^2 + 4)^2}$ has double poles at $z = \pm 2i$; only $2i$ is in the upper half-plane.

$$\operatorname{Res}_{z=2i} f = \frac{d}{dz} \left[\frac{1}{(z+2i)^2} \right]_{z=2i} = \frac{-2}{(4i)^3} = \frac{1}{32i}$$

On the large semicircle $|f| \sim 1/R^4$, so that part of the integral tends to zero. Therefore

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2+4)^2} = 2\pi i \cdot \frac{1}{32i} = \frac{\pi}{16}$$

Answer. $\pi/16 \approx 0.196$

5. Electrical and Electronics Engineering

Question 1: The Smith chart mapping

Question. Find the bilinear transformation that maps $z=0, 1, \infty$ to $w=-1, 0, 1$. Show that the normalised load-impedance plane ($\operatorname{Re} z \geq 0$) is mapped onto the unit disc, and explain how this gives the Smith chart reflection coefficient $\Gamma = \frac{z-1}{z+1}$.

Try $w = \frac{z-1}{z+1}$: $z=0 \mapsto -1$, $z=1 \mapsto 0$ and $z=\infty \mapsto 1$ ✓. A bilinear transformation is fixed uniquely by three points, so this is the map.

If $\operatorname{Re} z \geq 0$, then z is at least as close to 1 as to -1 , so $|z-1| \leq |z+1|$ and $|w| \leq 1$. The imaginary axis maps to the unit circle.

With $z = Z_L/Z_0$ as the normalised impedance, w is the reflection coefficient $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$. Every passive load ($\operatorname{Re} z \geq 0$) plots inside the unit disc; the Smith chart is that disc with the circles of constant resistance and constant reactance drawn on it.

Answer. $w = \frac{z-1}{z+1}$, which maps $\operatorname{Re} z \geq 0$ onto the unit disc.

Question 2: RLC impulse response

Question. An RLC circuit has transfer function $H(s) = \frac{1}{s^2 + 2s + 5}$. Find its poles, and classify them as simple or of higher order. Use Cauchy's integral formula or residues to find the impulse response $h(t)$.

$s^2 + 2s + 5 = 0$ gives two simple poles, $s = -1 \pm 2i$. Then

$$h(t) = \frac{e^{(-1+2i)t}}{4i} + \frac{e^{(-1-2i)t}}{-4i} = e^{-t} \frac{e^{2it} - e^{-2it}}{4i} = \frac{1}{2} e^{-t} \sin 2t$$

This is an underdamped response, and it is stable because the poles are in the left half-plane.

Answer. $h(t) = 1/2 e^{-t} \sin 2t$

6. Food Engineering

Question 1: Steady slab temperature

Question. The temperature in a long rectangular food slab in steady state is $T(x, y) = 20 + 5(x^3 - 3xy^2)$. Show that T is harmonic, and find the harmonic conjugate that gives the heat-flow lines. Write the analytic function $f(z)$ whose real part is $T - 20$.

Let $u = T - 20 = 5(x^3 - 3xy^2)$. Then $u_{xx} = 30x$ and $u_{yy} = -30x$, so $\nabla^2 u = 0$ and T is harmonic. From the Cauchy–Riemann equations:

$$v_y = u_x = 15x^2 - 15y^2 \Rightarrow v = 15x^2y - 5y^3 + g(x), v_x = -u_y = 30xy \Rightarrow g'(x) = 0$$

So $v = 5(3x^2y - y^3)$, whose level curves are the heat-flow lines, and

$$f(z) = u + iv = 5(x + iy)^3 = 5z^3$$

Answer. $v = 5(3x^2y - y^3)$; $f(z) = 5z^3$, so $T = 20 + \operatorname{Re}(5z^3)$.

Question 2: $\oint_C e^{2z}/(z-1)^3 dz$

Question. Evaluate $\oint_C \frac{e^{2z}}{(z-1)^3} dz$ where C is the circle of radius 2 centred at the origin, using Cauchy's integral formula for derivatives. Such integrals arise in inverting Laplace-domain models of thermal processing with repeated time constants.

The point $z = 1$ lies inside C . Using Cauchy's formula for derivatives, $\oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(a)$, with $f = e^{2z}$, $n = 2$ and $a = 1$, and $f''(z) = 4e^{2z}$:

$$\oint_C \frac{e^{2z}}{(z-1)^3} dz = \frac{2\pi i}{2!} \cdot 4e^2 = 4\pi e^2 i$$

Answer. $4\pi e^2 i \approx 92.9i$

7. Mechanical Engineering

Question 1: The Joukowski transformation

Question. The Joukowski transformation $w = z + \frac{1}{z}$ maps circles in the z -plane to aerofoil-like shapes. Find the singularities and the critical points (where $dw/dz = 0$) of the transformation, and show that the unit circle maps to the segment $[-2, 2]$ on the real axis.

The singularities are a simple pole at $z = 0$ and $z = \infty$. The critical points satisfy

$$\frac{dw}{dz} = 1 - \frac{1}{z^2} = 0 \Rightarrow z = \pm 1$$

The map is not conformal there; these points become the sharp trailing edge of the aerofoil. On the unit circle, $z = e^{i\theta}$ and

$$w = e^{i\theta} + e^{-i\theta} = 2 \cos \theta \in [-2, 2]$$

so the circle collapses onto a flat plate, covered twice. Shifting the circle's centre off the origin, while keeping it through $z = 1$, produces a cambered, thick aerofoil.

Answer. Pole at $z = 0$; critical points $z = \pm 1$; the unit circle maps to $[-2, 2]$.

Question 2: $\int_0^{2\pi} d\theta / (5 + 4 \cos \theta)$

Question. Evaluate $\int_0^{2\pi} \frac{d\theta}{5 + 4 \cos \theta}$ by contour integration round the unit circle. This type of integral appears in the analysis of rotating machinery with eccentric loading.

Let $z = e^{i\theta}$, so $\cos \theta = 1/2(z + z^{-1})$ and $d\theta = \frac{dz}{iz}$:

$$\int_0^{2\pi} \frac{d\theta}{5 + 4 \cos \theta} = \oint_{|z|=1} \frac{dz}{iz(5 + 2z + 2/z)} = \oint \frac{dz}{2i(z + 1/2)(z + 2)}$$

Only $z = -1/2$ is inside the circle, with residue $\frac{1}{2i \cdot 3/2} = \frac{1}{3i}$. So the integral is $2\pi i \cdot \frac{1}{3i} = \frac{2\pi}{3}$.

Answer. $2\pi/3 \approx 2.094$

8. Petroleum Engineering

Question 1: Injector-producer pair

Question. A source (injector) at $z = -a$ and a sink (producer) at $z = a$ give the complex potential $w = \frac{q}{2\pi} [\log(z + a) - \log(z - a)]$. Find the singularities of w and their type, and show that the equipotentials are circles (Apollonius circles). Find the stagnation points, if there are any.

w has **logarithmic branch points** at $z = -a$ (source) and $z = a$ (sink). These are not poles, because $\log$ is multivalued. The complex velocity

$$\frac{dw}{dz} = \frac{q}{2\pi} \left[\frac{1}{z+a} - \frac{1}{z-a} \right] = \frac{q}{2\pi} \cdot \frac{-2a}{z^2 - a^2}$$

has simple poles at $\pm a$. The equipotentials are $\phi = \frac{q}{2\pi} \ln \frac{|z+a|}{|z-a|} = \text{const}$, that is $\frac{|z+a|}{|z-a|} = k$: the

Apollonius circles, which have $\pm a$ as inverse points. dw/dz is never zero for finite z , so there are no stagnation points. All the fluid injected at $-a$ goes to the producer at a .

Answer. Logarithmic branch points at $\pm a$; equipotentials are Apollonius circles; no stagnation points.

Question 2: Mapping the off-centre well to the centre

Question. Find the bilinear transformation that maps the circle of radius R centred at the origin (the reservoir boundary) to the unit circle, with an off-centre well at $z = z_0$ going to the centre $w = 0$. Explain how this lets the known solution for a centred well be used for an off-centre well.

$$w = \frac{R(z - z_0)}{R^2 - \bar{z}_0 z}$$

- At $z = z_0$, $w = 0$ ✓.
- On $|z| = R$, $z\bar{z} = R^2$, so $|R^2 - \bar{z}_0 z| = |z||\bar{z} - \bar{z}_0| = R|z - z_0|$, hence $|w| = 1$ ✓.
- $z = 0$ maps to $-z_0/R$, which is inside the unit disc, so the inside maps to the inside.

The pressure for a *centred* well in a circular reservoir is known: $p = p_w + \frac{q\mu}{2\pi k h} \ln \frac{r}{r_w}$. Because the map is conformal, Laplace's equation is preserved. Taking that solution in the w -plane and substituting $w(z)$ gives the pressure field of the off-centre well directly, and also its effective shape factor for productivity calculations.

Answer. $w = \frac{R(z - z_0)}{R^2 - \bar{z}_0 z}$